

EEE 312 · DIGITAL SIGNAL PROCESSING I LABORATORY · BUET

Voice-Based Meal Verification System for Hall Dining

Complete project guide, from the basics to the viva: theory, mathematics, diagrams, code, outputs and every improvement in **v4.1.4_claude**

Samin Zahan Khan

2206135

Syed Ibnul Morshed

2206141

S M Sindid Islam**Mahodi**

2206147

Abdullah Al Rohan

2206150

Group 7 (Section C1) · Department of EEE · Version of the code described: **Final project v4.1.4_claude**, engine 1.1 (30 September 2026)

How to use this guide

Every chapter has the same rhythm: the **idea** in one paragraph, the **mathematics** in a shaded box, the **MATLAB code** that implements it (copied from the project files), an **"In plain words"** explanation, and a **"What would happen if..."** box that shows why each design choice was made. The last chapters contain the complete evaluation, the A-to-Z list of improvements and more than 70 viva questions with answers.

If you only have ten minutes before the viva, read Chapter 1, Chapter 10 (why v4.1.4 failed), Chapter 11 (results) and Chapter 15 (viva).

Contents

1. The project in one chapter

- 1.1 The problem
- 1.2 Where the project came from
- 1.3 The solution at a glance
- 1.4 Headline results (all voices in the project folder)
- 1.5 Quick start

2. Basics you need first

- 2.1 Sound as a discrete-time signal
- 2.2 How speech is produced: the source-filter model
- 2.3 DFT, FFT and the spectrum
- 2.4 Short-time analysis and windows
- 2.5 Digital filters, the z-transform and pole-zero plots
- 2.6 A little probability: Gaussians, likelihood and the log-likelihood ratio
- 2.7 Error rates used to judge a biometric system

3. The front-end: from microphone samples to features (vsd_frontend.m)

- 3.1 Rational resampling to 8 kHz
- 3.2 DC removal and spectral subtraction
- 3.3 Peak normalisation and pre-emphasis
- 3.4 Framing, power spectrum, mel filter bank, log and DCT (MFCC)
- 3.5 Speech mask and deltas
- 3.6 CMVN: the mathematical reason the system survives a microphone change

4. Matching the words: DTW and cohort scores

- 4.1 Dynamic Time Warping
- 4.2 Multi-template matching and cohort normalisation

5. Who is speaking: GMM-UBM voice biometrics (vsd_gmm.m)

- 5.1 Gaussian mixture model
- 5.2 Universal Background Model trained by EM
- 5.3 MAP adaptation: one model per student from 10–20 s of speech
- 5.4 The voice score

6. The decision: fused score, imposter identification, unknown speaker (vsd_decide.m)

6.1 The rule

6.2 Why these weights and thresholds

6.3 Worked example 1: the student v4.1.4 kept rejecting (2206147)

6.4 Worked example 2: an imposter is named

6.5 v1.1: the "clear winner" rule, and why we did not just lower the threshold

6.6 Field log: an imposter the old matcher let in, and how v4.1.4_claude stops it

7. Replay-attack prevention: three layers

7.1 Layer 1: the random-digit challenge (vsd_challenge.m)

7.2 Layer 2: replay memory, the digital-copy detector (vsd_replay_guard.m)

7.3 Layer 3: the proposal's sub-200 Hz liveness heuristic (advisory)

8. Enrolment, the model cache and self-adaptation

8.1 Building the models (vsd_build_models.m, vsd_models.m)

8.2 Self-adaptation to a new microphone (vsd_adapt.m)

8.3 Recording the digits

9. The complete transaction, the GUI and the dining rules

9.1 One counter transaction (vsd_verify_transaction.m + verify_meal_workflow.m)

9.2 What the operator sees: the six gates

9.3 Logs

9.4 Dining business logic (kept from v4.1.4)

10. Debugging v4.1.4: why genuine students were rejected, and what was changed

10.1 The evidence

10.2 Root causes found in the code

10.3 File-by-file changelog

11. Evaluation: protocols, outputs and how to read them

11.1 The data

11.2 Protocols

11.3 Results

11.4 Demonstration output (RUN_ME('demo'))

11.5 Transaction sweep: one full transaction per student

11.6 Output files

12. A to Z: every improvement, and why**13. "What would happen if...": consolidated scenarios****14. Limitations and future work**

14.1 Limitations

14.2 Future work

15. Viva questions and answers

A. Signals and sampling

B. Features

C. DTW and cohort scores

D. GMM-UBM

E. Decision and imposter identification

F. Replay attacks

G. Evaluation

H. Code and MATLAB

I. System, ethics and outcomes

16. Appendix

16.1 Every parameter (vsd_config.m)

16.2 Where things are

16.3 Troubleshooting

16.4 Glossary

16.5 References

17. Master DSP Handbook, Time Distance & Code Tuning Guide

17.1 Comprehensive DSP Lexicon: Terms, Mathematical Formulations, Purpose & Quantitative Impact

17.2 Time Distance & Dynamic Time Warping (DTW) Demystified

17.3 Code Tuning Handbook: Exact File Locations, Line Numbers & Tuning Instructions

17.4 Step-by-Step Numerical Calculations Demonstrating System Improvements

1. The project in one chapter

1.1 The problem

BUET residential halls hand out meals against paper meal cards. Cards are shared, lost or sold to non-residents, and the queue moves slowly. We replace the card with the student's **voice**. At the counter the student says their **Student ID (roll number)** and their **full name**. The system must answer four questions:

1. **Who was named?** Which of the enrolled students do the spoken words refer to, or nobody?
2. **Is it really them?** Does the voice belong to that student, or is a friend speaking for them (an *imposter*, a "proxy meal")? If it is a friend, who is it?
3. **Is it live?** Is a person speaking now, or is a phone playing a *recording* of the student (a *replay attack*)?
4. **May they eat?** Is the student on this month's paid roster, is a meal window (breakfast, lunch, dinner) open, and have they not already eaten in this window?

1.2 Where the project came from

Stage	What it was	Main weakness
Senior baseline (Group 1, EEE'21, "Voice Based Secured Residential Hall Entrance")	MATLAB script. Fixed 3 s recording at 44.1 kHz, pre-emphasis 0.99, block-energy silence removal, 12 MFCC + Δ + $\Delta\Delta$, one 8-codeword VQ codebook (LBG) per phrase. Access if the nearest ID codebook and the nearest name codebook belong to the same person.	No rejection threshold: anyone saying a name was accepted as the nearest student. No noise or channel handling, no anti-spoofing, no meal logic.
Our proposal (24 July 2026)	Five modifications: AGC + rational resampling, sub-200 Hz liveness, spectral subtraction, STE+ZCR endpointing, dining business logic with CSV logs.	These are sound DSP ideas, but they do not decide identity robustly.
v4.1.4 (our previous release)	GUI kiosk with 8 kHz processing, spectral subtraction, DTW matching, monthly roster, meal windows, admin tools.	Rejected most genuine students in live use (48 of 65 logged attempts), especially on a new microphone. See Chapter 10.
v4.1.4_claude (this guide)	Same GUI and business logic, plus a new verification engine (files <code>v4d_*.m</code>). It uses two independent kinds of evidence, names the imposter and has three anti-replay layers.	Limitations are listed honestly in Chapter 14.

1.3 The solution at a glance

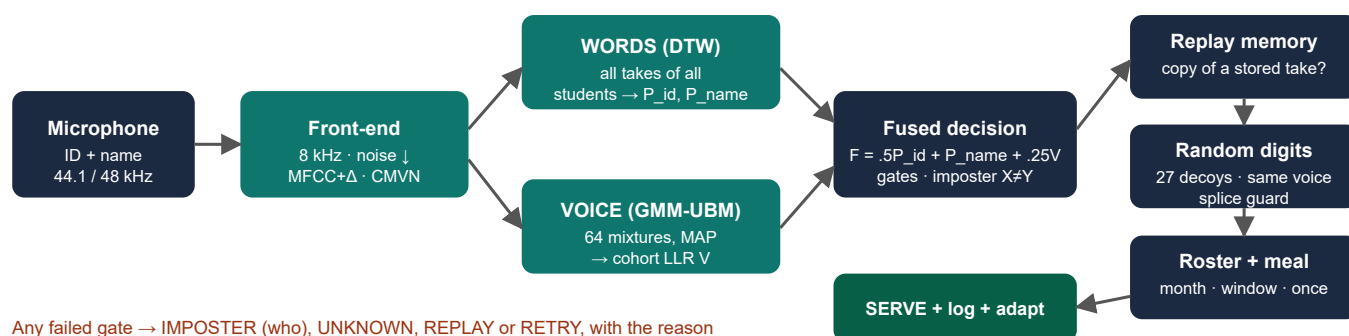


Figure 1. the v4.1.4_claude pipeline. Two independent kinds of evidence (words and voice) are fused, then the freshness checks run.

Two independent questions are asked of the same two recordings:

- **WHAT was said** (text-dependent). The spoken ID and name are compared with *every stored take of every student* by Dynamic Time Warping (DTW). The distances are turned into *cohort scores* `P_id` and `P_name` : "how much closer is this recording to student k than to k's closest rivals?"
- **WHO is speaking** (text-independent). All speech frames are scored by a Gaussian Mixture Model voice model of every student, adapted from a Universal Background Model (GMM-UBM). The log-likelihood ratios become a cohort-normalised voice score `v` .

A fused score picks the candidate. Gates then check each kind of evidence separately. If the words point to student X but the voice clearly belongs to another enrolled student Y, the attempt is refused as an **IMPOSTER** and Y is reported. Before a meal is served the student must also say **three random digits** shown on screen. A recording made earlier cannot contain digits chosen a second ago.

1.4 Headline results (all voices in the project folder)

98.3 %

genuine students accepted,
leave-one-take-out, all
microphones (114/116)

90.0 %

genuine accepted when
enrolled on one mic and
tested on others (18/20)

0 / 116

unknown (not enrolled)
speakers accepted

0.22 %

imposters with perfect words
accepted (8 of 3596); the
imposter is named in 97.4 %

Under the protocol where all systems are enrolled with take 1 only, the senior baseline accepted **25.9 %** of genuine students on a different microphone and v4.1.4 accepted **7.4 %**. v4.1.4_claude 1.1 accepts **85.2 %**, while unknown speakers fall from 15.4 % (senior) to 0.9 %. Every replayed, stale or spliced recording in the demonstration was refused.

1.5 Quick start

```
>> cd 'Final project v4.1.4_claude'
>> RUN_ME                % counter GUI (first run builds the voice models, ~1 min)
% the status box must start with "Engine: v4.1.4_claude-1.1"; otherwise the old folder is running
>> RUN_ME('demo')        % 7 transactions: genuine, 2 imposters, 4 replay attacks
>> RUN_ME('evaluate')    % every stored voice: LOTO, cross-mic, unknown speaker, mimic
>> RUN_ME('rebuild')     % re-extract every recording (after changing vsd_config)
```

The recordings are read from `VSD_Enrollment/Train` . If that folder is missing, the sibling folder `../Final project v4.1.4/VSD_Enrollment/Train` is used instead. The folder contains:

- `ID/<7-digit id>/*.wav` : spoken roll number takes. **All** takes are used.
- `Name/<id>/*.wav` : spoken full-name takes.
- `Digits/<id>/<id>_<digit>_01.wav` : isolated digits zero...nine.
- `ID/<id>/Adaptive/` and `Name/<id>/Adaptive/` : takes learnt automatically on new microphones.

2. Basics you need first

2.1 Sound as a discrete-time signal

A microphone turns air pressure into a voltage $x(t)$. The sound card **samples** it every $T_s = 1/f_s$ seconds and **quantises** each sample, for example to 16 bits. MATLAB then holds a vector $x[n] = x(nT_s)$. Our laptops capture at $f_s = 44.1$ kHz, the Group-2 microphone at 48 kHz, and the engine works at 8 kHz.

Sampling theorem (Nyquist–Shannon): $x(t)$ is recoverable from $x[n]$ if it contains no energy at or above $f_s/2$.
8 kHz sampling $\rightarrow$ band 0 ... 4 kHz. Speech formants F1–F3 ($\approx$ 300–3500 Hz) fit inside.

In plain words

Sampling is taking photographs of the wave. If you take them too slowly, a fast wave looks like a slow one (*aliasing*). 8000 photos per second are enough for the part of the voice that carries the words and most of the speaker's character, and this is 5.5 $\times$ less data than 44 100.

2.2 How speech is produced: the source–filter model

The vocal folds make a buzz (voiced sounds, with fundamental frequency $F_0 \approx 85$ –180 Hz for men and 165–255 Hz for women) or turbulent noise (unvoiced sounds such as *s*, *f*). The vocal tract (throat, mouth, nose) acts as a filter whose resonances are the **formants**. In the frequency domain:

$$S(\omega) = G(\omega) \cdot V(\omega) \cdot R(\omega) \text{ (glottal source} \times \text{vocal-tract filter} \times \text{lip radiation)}$$

$$\text{What the computer receives: } Y(\omega) = S(\omega) \cdot H_{\text{mic+room}}(\omega) + N(\omega)$$

The **words** are mostly in how $V(\omega)$ changes over time. The **speaker** is in the average shape of $V(\omega)$ (vocal-tract length) and in $G(\omega)$. The **microphone and room** add $H(\omega)$, and the **hall** adds noise $N(\omega)$. Every block of our front-end is there to keep V and G while removing H and N .

2.3 DFT, FFT and the spectrum

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi kn/N}, k = 0 \dots N-1, \text{ frequency of bin } k: f_k = k \cdot f_s/N$$

$$\text{Power spectrum } P[k] = |X[k]|^2 / N \cdot \text{dB: } 10 \cdot \log_{10}(P)$$

The FFT computes the DFT in $O(N \log N)$. We use $N_{\text{FFT}} = 256$ at 8 kHz, so each bin is 31.25 Hz wide. Only bins 0...128 (0–4 kHz) are unique for a real signal.

2.4 Short-time analysis and windows

Speech is **non-stationary**, but over 20–30 ms the vocal tract hardly moves. We therefore cut the signal into 25 ms **frames** (200 samples) every 10 ms (80 samples) and take one spectrum per frame. Cutting abruptly (a rectangular window) smears energy into every bin (*spectral leakage*). A **Hamming window** tapers the edges:

$$w[n] = 0.54 - 0.46 \cdot \cos(2\pi n/(N-1)), n = 0 \dots N-1$$

first side-lobe: rectangular –13 dB, Hamming –43 dB; main-lobe width: rectangular $4\pi/N$, Hamming $8\pi/N$

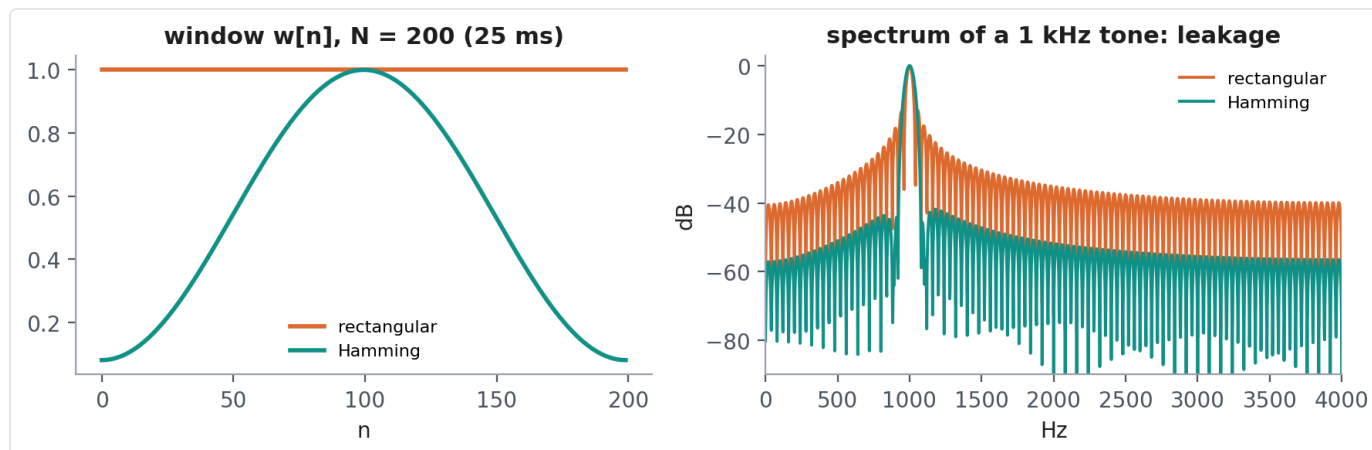


Figure 2. Rectangular vs Hamming window on a 1 kHz tone. Hamming pays with a wider main lobe but its side lobes are 30 dB lower, so formants of neighbouring frequencies stay separate.

2.5 Digital filters, the z-transform and pole-zero plots

FIR: $y[n] = \sum_k b_k x[n-k] \Leftrightarrow H(z) = \sum_k b_k z^{-k}$ (only zeros, always stable, can be linear-phase)

IIR: $y[n] = \sum b_k x[n-k] - \sum a_k y[n-k] \Leftrightarrow H(z) = B(z)/A(z)$ (poles must lie inside $|z| = 1$)

Frequency response: $H(e^{j\omega}) = H(z)|_{z=e^{j\omega}}$, $\omega = 2\pi f/f_s$

The project uses FIR filters throughout: the pre-emphasis filter $1 - 0.97z^{-1}$, the delta (differentiator) filter, the polyphase anti-alias filter inside `resample`, and the loudspeaker simulator's Butterworth band-pass in the tests.

2.6 A little probability: Gaussians, likelihood and the log-likelihood ratio

$N(x; \mu, \sigma^2) = (2\pi\sigma^2)^{-1/2} \exp(-(x-\mu)^2/(2\sigma^2))$; diagonal d-dimensional: product of d such terms

Likelihood of frames $x_{1...T}$ under a model λ (frames assumed independent): $p(X|\lambda) = \prod_t p(x_t|\lambda)$

Log-likelihood ratio: $LLR = (1/T) \sum_t [\ln p(x_t|\text{student}) - \ln p(x_t|\text{everyone})]$

In plain words

A likelihood says how well a model explains the data. On its own it is meaningless, because a long or noisy recording makes every likelihood small. A **ratio** against a reference ("people in general") cancels those shared effects, so $LLR > 0$ simply means "this sounds more like the student than like an average person on this microphone".

2.7 Error rates used to judge a biometric system

Term	Meaning	Our protocol
GAR (genuine acceptance rate) = $1 - \text{FRR}$	Fraction of genuine attempts accepted as the right student	LOTO, cross-mic, transaction sweep
FRR (false rejection rate)	Genuine attempts refused. This is what annoyed students with v4.1.4.	$1 - \text{GAR}$
FAR (false acceptance rate)	Attempts by a non-owner that are accepted	Unknown speaker, perfect mimic
Wrong-student accept	A genuine student accepted as <i>someone else</i>	Reported for every genuine protocol
EER	Threshold where FAR = FRR (one-number summary)	The FAR/FRR figure in §6.2

3. The front-end: from microphone samples to features (vsd_frontend.m)

Every recording is processed once by $u = \text{vsd_frontend}(x, f_s, c)$. This applies to live captures, stored templates, digits and challenge answers alike, so all comparisons are made on identical terms. The output u contains `Content` (features for the word match), `Speaker` (features for the voice model), `SpeechSec`, `RawRms`, `ClipFrac` and the 8 kHz audio.



Figure 3. the front-end chain (vsd_frontend.m). Each block is explained in §3.1–3.6.

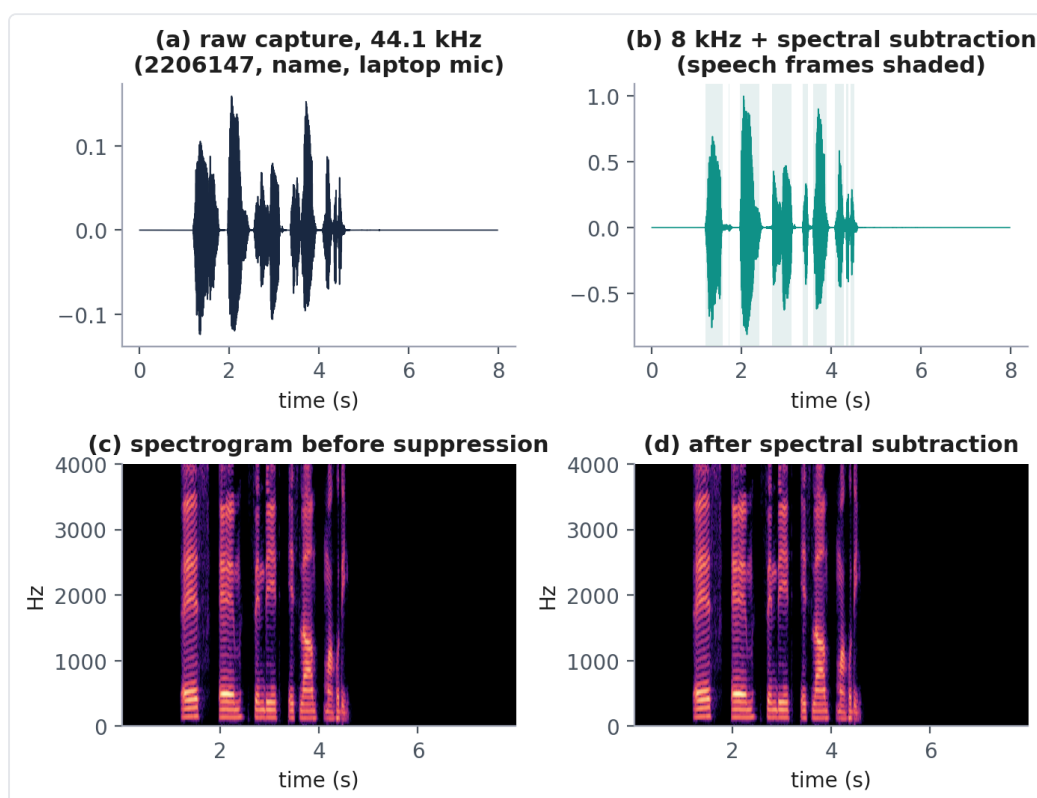


Figure 4. The real signal chain on 2206147's name (take 4, laptop microphone, 44.1 kHz). (a) raw capture; (b) after resampling to 8 kHz and spectral subtraction, with the frames kept by the speech mask shaded; (c)/(d) spectrograms before and after noise suppression. The background between words is darker in (d).

3.1 Rational resampling to 8 kHz

44.1 kHz → 8 kHz: $8000/44100 = 80/441 = L/M$ (gcd = 100) 48 kHz → 8 kHz: 1/6

Up-sample by L (insert $L-1$ zeros) → low-pass $H(e^{j\omega})$ with cut-off $\omega_c = \pi/\max(L, M)$ and gain L → keep every M -th sample

Polyphase implementation: only the output samples that are kept are ever computed (cost $\propto$ output length)

```
function y = vsd_resample(x, fsIn, fsOut)
if fsIn == fsOut, y = x; return; end
g = gcd(round(fsOut), round(fsIn));
p = round(fsOut)/g; q = round(fsIn)/g; % 80 and 441 for a laptop mic
y = resample(x, p, q); % polyphase FIR, Kaiser window
y = y(:);
```

In plain words

We need to go from 44 100 to 8 000 samples per second. This is not a whole-number ratio, so we "stretch by 80, smooth, keep 1 in 441". The smoothing low-pass filter removes everything above 4 kHz first. Without it, a hiss at 6 kHz would fold down and appear as a fake 2 kHz component.

What would happen if...

Naive decimation ($x(1:5:end)$, about 8.8 kHz with no filter): energy between 4.4 and 22 kHz aliases into the speech band, and fricatives such as *s* in "Sindid" turn into broadband noise over the formants. **$f_s = 4$ kHz:** the band ends at 2 kHz, so F3 and much of F2 are lost, and names such as "Samin" and "Sindid" become hard to tell apart. **Staying at 44.1 kHz:** 5.5× more samples to process, and 5.5× longer frames in samples, with no gain for 26 mel bands below 4 kHz.

3.2 DC removal and spectral subtraction

Speech and hall noise add: $y = s + n$. The speech is non-stationary, while fans, crowd hum and hiss change slowly. So we estimate the noise power spectrum and subtract it:

Frames: 32 ms periodic Hann, 50 % overlap-add (the Hann windows sum to 1). $Y_m[k] = \text{FFT of frame } m$

Noise estimate: $\hat{P}_N[k] = \text{mean of } |Y_m[k]|^2 \text{ over the 10 \% of frames with the least total energy}$

Gain: $G_m[k] = \max(1 - \alpha \cdot \hat{P}_N[k] / |Y_m[k]|^2, \beta)$, $\alpha = 2$ (over-subtraction), $\beta = 0.02$ (floor)

$\hat{S}_m[k] = \sqrt{G_m[k]} \cdot Y_m[k]$ (phase kept) $\rightarrow$ inverse FFT $\rightarrow$ overlap-add

```
[~, order] = sort(sum(Pw, 1)); % frames from quietest to loudest
q = order(1:max(3, round(c.SSNoiseFrac * nF))); % quietest 10 %
Pn = mean(Pw(:, q), 2); % noise power per bin
G = max(1 - c.SSAlpha * Pn ./ max(Pw, 1e-20), c.SSBeta);
yf = real(ifft(F .* sqrt(G), N, 1)); % then overlap-add
```

In plain words

Every recording contains a little silence. The quietest tenth of the recording tells us what the room sounds like. We remove twice that much from every frame ($\alpha = 2$), but never more than 98 % of a frame's power ($\beta = 0.02$ keeps a faint floor so no "musical noise" chirps appear). The proposal's 0.5 s pre-roll idea is kept for the live capture, and this version also works on stored files that have no pre-roll.

What would happen if...

$\alpha = 1$: residual noise stays in the pauses, so the speech mask keeps more non-speech frames. **$\alpha = 4$:** weak consonants are deleted along with the noise. **$\beta = 0$:** isolated spectral peaks survive between gaps as "musical noise" that DTW treats as content. **The noise estimated only from a fixed first 0.5 s:** if a student starts talking early, the "noise" is speech and the first word is cancelled. This happened with some archived takes.

3.3 Peak normalisation and pre-emphasis

$y \leftarrow y / \max|y|$ (level independence, the proposal's AGC idea; CMVN later removes what remains)

Pre-emphasis: $y_p[n] = y[n] - 0.97 \cdot y[n-1] \Leftrightarrow H(z) = 1 - 0.97 z^{-1}$ (zero at $z = 0.97$, pole at $z = 0$)

$|H(e^{j\omega})|^2 = 1 + 0.97^2 - 2 \cdot 0.97 \cdot \cos \omega \rightarrow -30.5 \text{ dB at DC, } +5.9 \text{ dB at 4 kHz } (\approx +6 \text{ dB/octave over the speech band})$

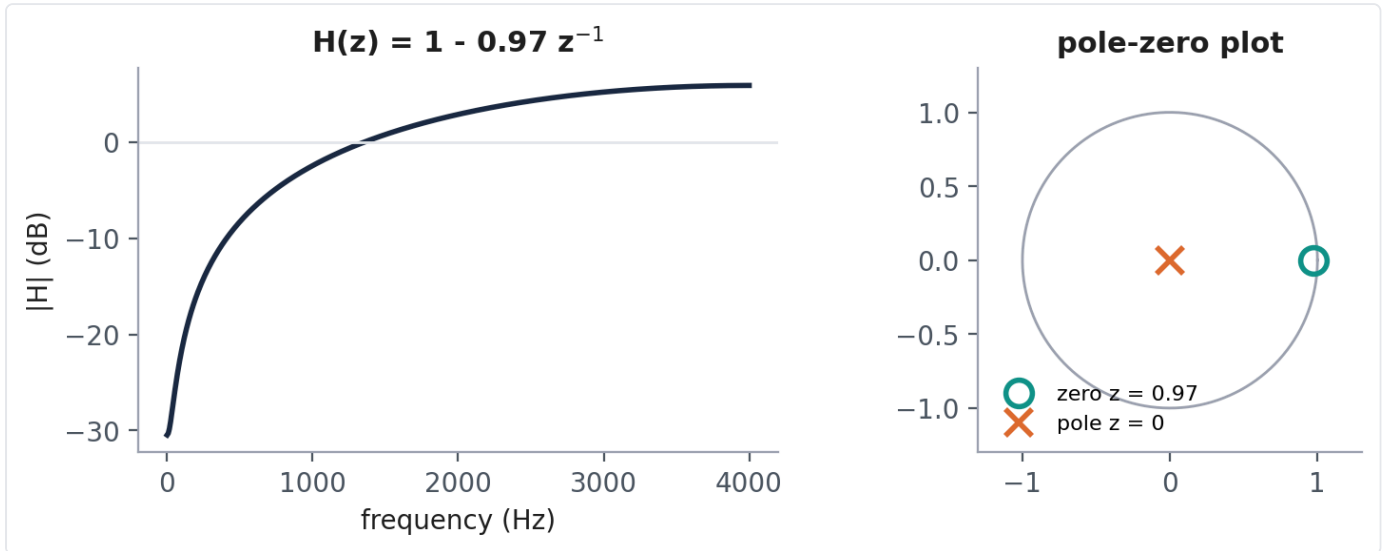


Figure 5. Frequency response and pole-zero plot of the pre-emphasis filter. The senior code used 0.99, which cuts the low band even harder (-40 dB at DC).

```
y = [y(1); y(2:end) - c.PreEmph * y(1:end-1)]; % c.PreEmph = 0.97
```

What would happen if...

No pre-emphasis: the spectral tilt of voiced speech (-6 dB/oct) lets F1 dominate every mel band. F3 and F4, which carry vocal-tract length (speaker) and place of articulation (words), are weighted about 10–20 dB lower in the cepstrum. **0.99 instead of 0.97:** nearly identical above 300 Hz. The difference is below 200 Hz, which is why the proposal's sub-200 Hz liveness test must use the raw native-rate signal.

3.4 Framing, power spectrum, mel filter bank, log and DCT (MFCC)

Frames: $N = 200$ samples (25 ms), hop $H = 80$ (10 ms), Hamming window, $N_{\text{FFT}} = 256 \rightarrow P_t[k] = |X_t[k]|^2/N$

Mel scale: $\text{mel}(f) = 2595 \cdot \log_{10}(1 + f/700)$. 26 triangles with centres equally spaced in mel between 100 and 3800 Hz

Filter-bank energies: $E_t[m] = \sum_k B_m[k] \cdot P_t[k]$, $m = 1 \dots 26$

Cepstrum (DCT-II): $c_t[n] = \sqrt{(2/M) \sum_{m=1}^M \ln E_t[m] \cdot \cos(\pi n(m - 1/2)/M)}$, $n = 0 \dots 12$

Lifter: $c_t[n] \leftarrow c_t[n] \cdot (1 + (L/2) \cdot \sin(\pi n/L))$, $L = 22$. c_0 (overall loudness) is discarded; $c_1 \dots c_{12}$ are kept

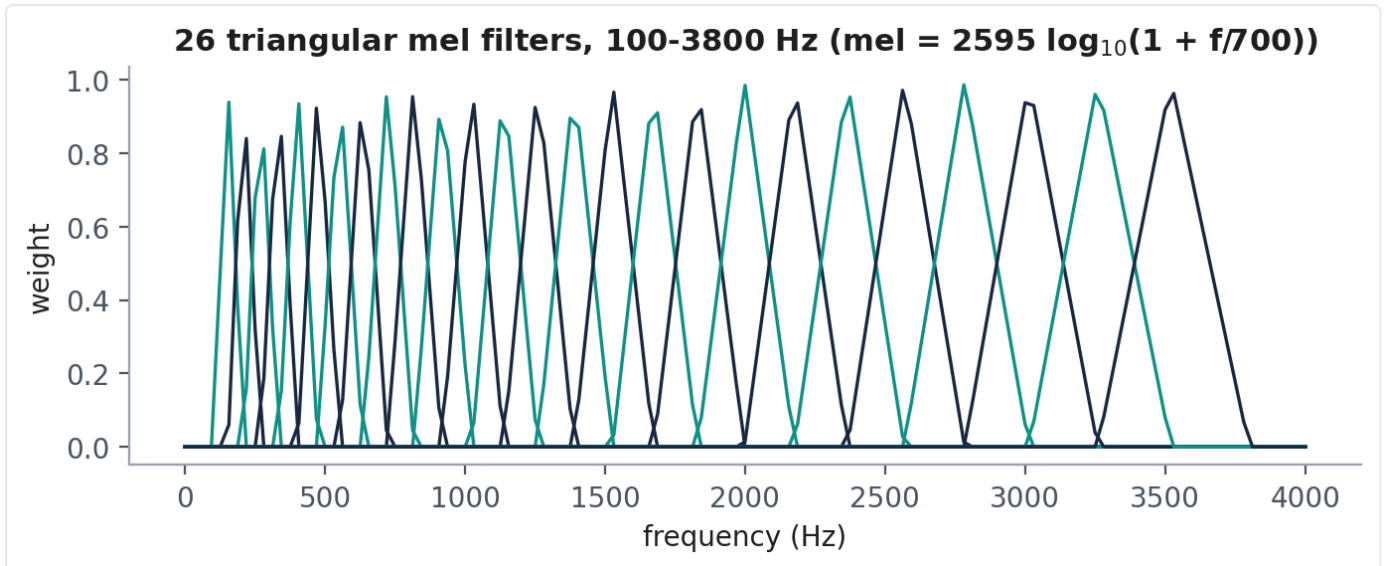


Figure 6. The 26 mel filters. They are narrow at low frequencies, where the ear (and F1/F2) needs detail, and wide at high frequencies.

```
idx = (1:N) + H * (0:nf-1)'; % [nf x N] sample indices of every frame
w = 0.54 - 0.46 * cos(2*pi*(0:N-1)/(N-1)); % Hamming
fr = y(idx) .* w;
S = fft(fr, c.NFFT, 2);
P = abs(S(:, 1:c.NFFT/2+1)).^2 / N; % power spectrum, 129 bins
B = vsd_melbank(c.NumMel, c.NFFT, c.Fs, c.MelLowHz, c.MelHighHz);
me = P * B.'; % 26 band energies per frame
logm = log(max(me, 1e-12));
cc = logm * vsd_dct(c.NumMel, c.NumCep).'; % c0..c12
cc = cc .* (1 + (c.Lifter/2) * sin(pi * (0:c.NumCep-1) / c.Lifter));
```

In plain words

For each 25 ms slice: measure how much energy falls into 26 ear-like bands, take the logarithm (as loudness perception does), then describe the band pattern with 12 numbers (the DCT is a compact "shape code"). Low cepstral coefficients describe the broad envelope, which is the vocal tract. The log is the key step: it turns multiplication by the microphone response into *addition* (3.6).

What would happen if...

Filters up to 7000 Hz at 8 kHz (the senior setting at 44.1 kHz): above 4 kHz there is nothing at 8 kHz, so the empty bands produce $\ln(0)$ and garbage. The upper edge must stay below Nyquist, which is why we use 3800 Hz. **Keeping c0:** the features then encode how loud the student spoke, so standing closer to the mic changes the match. **20 cepstra instead of 12:** tried and measured. Cross-mic acceptance fell from 78.3 % to 52.2 %, because the high cepstra encode fine spectral detail (pitch harmonics, microphone ripple) rather than the vocal tract.

3.5 Speech mask and deltas

Frame log-energy $e_t = \ln \sum_m E_t[m]$; keep frame t if $e_t > \max_t e_t - 3.0 \cdot \ln 10$ (i.e. within 30 dB of the loudest frame)

Delta (regression over ± 2 frames): $\Delta c_t = \sum_{n=1}^2 n \cdot (c_{t+n} - c_{t-n}) / (2 \cdot \sum n^2) = [2(c_{t+2} - c_{t-2}) + (c_{t+1} - c_{t-1})] / 10$

The delta is an FIR filter along time with impulse response $[2, 1, 0, -1, -2]/10$, a smoothed differentiator. Its DC gain is 0, so any constant offset (such as a microphone's cepstral offset) disappears from Δc .

What would happen if...

No speech mask: 2–5 s of silence around a 1.3 s name dominate the DTW path, and every recording "matches" every other in the silent parts. **A 60 dB floor:** breath and room noise are kept. **A 10 dB floor:** weak final consonants ("-d" in Sindid) are cut. The senior code's block silence remover (4096-sample blocks kept only if above the mean energy) cuts at block edges, often in the middle of a syllable.

3.6 CMVN: the mathematical reason the system survives a microphone change

Microphone + room = LTI filter: $Y_t(\omega) = X_t(\omega) \cdot H(\omega)$

Filter-bank energies (H nearly constant inside one band): $E_t^Y[m] \approx |H_m|^2 \cdot E_t^X[m]$

Log: $\ln E_t^Y[m] = \ln E_t^X[m] + \ln |H_m|^2 \rightarrow$ DCT is linear: $c_t^Y = c_t^X + h$, h does not depend on t

Cepstral mean subtraction: $c_t^Y - (1/T)\sum_{\tau} c_{\tau}^Y = c_t^X + h - ((1/T)\sum_{\tau} c_{\tau}^X + h) = c_t^X - \text{mean}(c^X)$ ✓ h cancels exactly

Variance normalisation: divide each dimension by its standard deviation $\rightarrow$ also cancels a gain on the dynamics

```
st = cc(:, 2:end); % c1..c12
d1 = vsd_delta(st, c.DeltaN); % deltas
cs = st(speech, :); ds = d1(speech, :); % speech frames only
content = [cs ds]; % 24-D word features
mu = mean(content, 1); sd = std(content, 1, 1);
U.Content = (content - mu) ./ sd; % CMVN
U.Speaker = [(cs - mean(cs,1)) ./ std(cs,1,1), ds ./ std(ds,1,1)]; % voice features
```

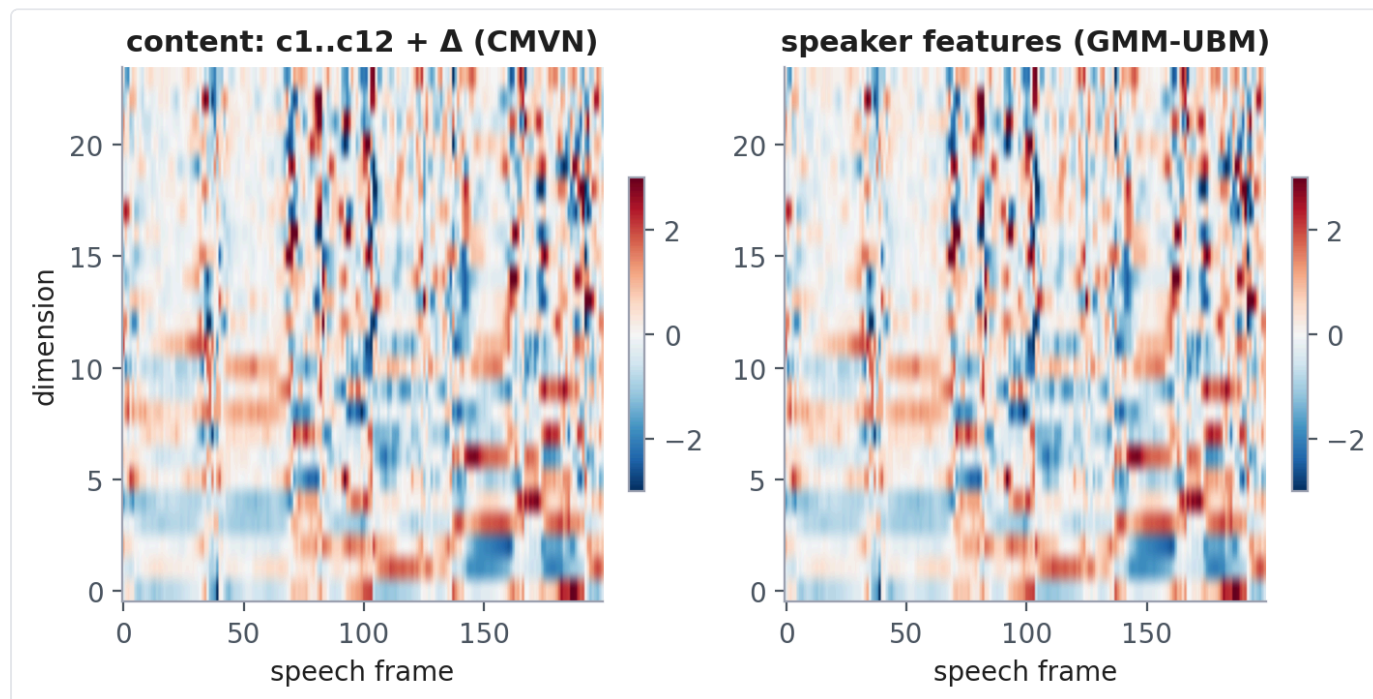


Figure 7. The two feature matrices for 2206147's name. Each is zero-mean and unit-variance per utterance, whichever microphone recorded it.

Watch out

v4.1.4 had cepstral mean subtraction **switched off** by its frozen calibration file (`mfcc.CmsNormalise = 0`). The microphone term h then stayed inside every frame, so a new microphone raised the distance to every template at once. This is root cause 2 in Chapter 10.

What would happen if...

CMS only, no variance normalisation, on the voice features: measured cross-microphone voice identification falls to 60.9 % (with CMVN: 78.3–91.3 %). **CMVN over a very short utterance** (< 0.3 s): the mean itself becomes content-dependent. This is why a minimum of 0.35 s of speech per phrase is enforced (c.MinSpeechSec).

4. Matching the words: DTW and cohort scores

4.1 Dynamic Time Warping

The same student never says "two two zero six one four seven" at the same speed twice. DTW finds the best monotone alignment between the query frames $a_{1\dots n}$ and the template frames $b_{1\dots m}$:

local cost $d(i, j) = \|a_i - b_j\|_2$ (24-D Euclidean)

$D(0,0) = 0$, $D(i, j) = d(i, j) + \min\{D(i-1, j-1), D(i-1, j), D(i, j-1)\}$

Sakoe–Chiba band: only $|i - j| \leq w$, $w = \max(\lceil 0.4 \cdot \max(n, m) \rceil, |n - m| + 1)$

distance = $D(n, m) / K$, K = number of cells on the optimal path (average mismatch per aligned frame pair)

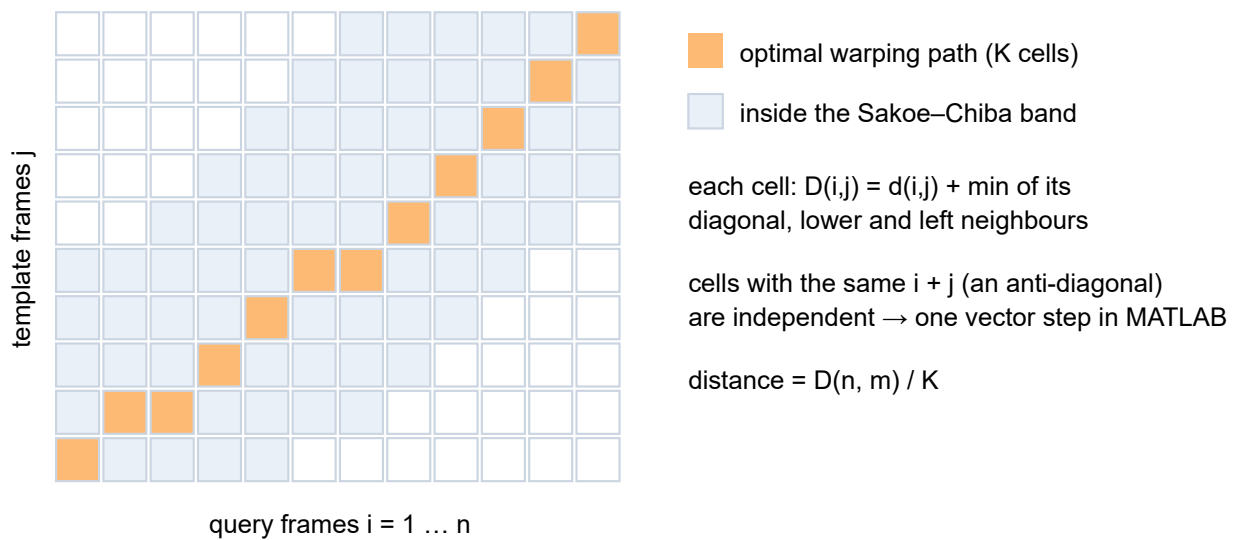


Figure 8. the DTW grid, the band and the warping path.

v4.1.4 divided by $n + m$. Dividing by the real path length K makes the distance the *mean frame mismatch*, so a 1 s name and a 3 s roll number are on the same scale. Ties are broken in favour of the diagonal step, as in the Python prototype.

Fast implementation: one anti-diagonal at a time (vsd_dtw.m)

```
C = sqrt(max(sum(A.^2,2) + sum(B.^2,2).' - 2*(A*B.'), 0)); % all local costs at once
for s = 2:(n + m) % cells with i + j = s
    iLo = max([1, s - m, ceil((s - bw)/2)]); % stay inside the band
    iHi = min([n, s - 1, floor((s + bw)/2)]);
    I = (iLo:iHi)'; J = s - I;
    x1 = D(kD); x2 = D(kU); x3 = D(kL); % diagonal, up, left predecessors
    useD = (x1 <= x2) & (x1 <= x3); useU = ~useD & (x2 <= x3);
    best = x3; best(useU) = x2(useU); best(useD) = x1(useD);
    D(kC) = C(I + (J - 1) * n) + best; % whole diagonal in one vector step
    Lp(kC) = lp + 1; % path length bookkeeping
end
dist = D(R, m + 1) / max(pathLen, 1);
```

In plain words

Every cell on the anti-diagonal $i + j = s$ depends only on the two previous anti-diagonals. So MATLAB can fill a whole anti-diagonal in one vector operation instead of a double for loop. There are $n + m$ interpreted steps instead of $n \cdot m$, with an identical result (checked against the Python prototype: 6.3311052297 in both).

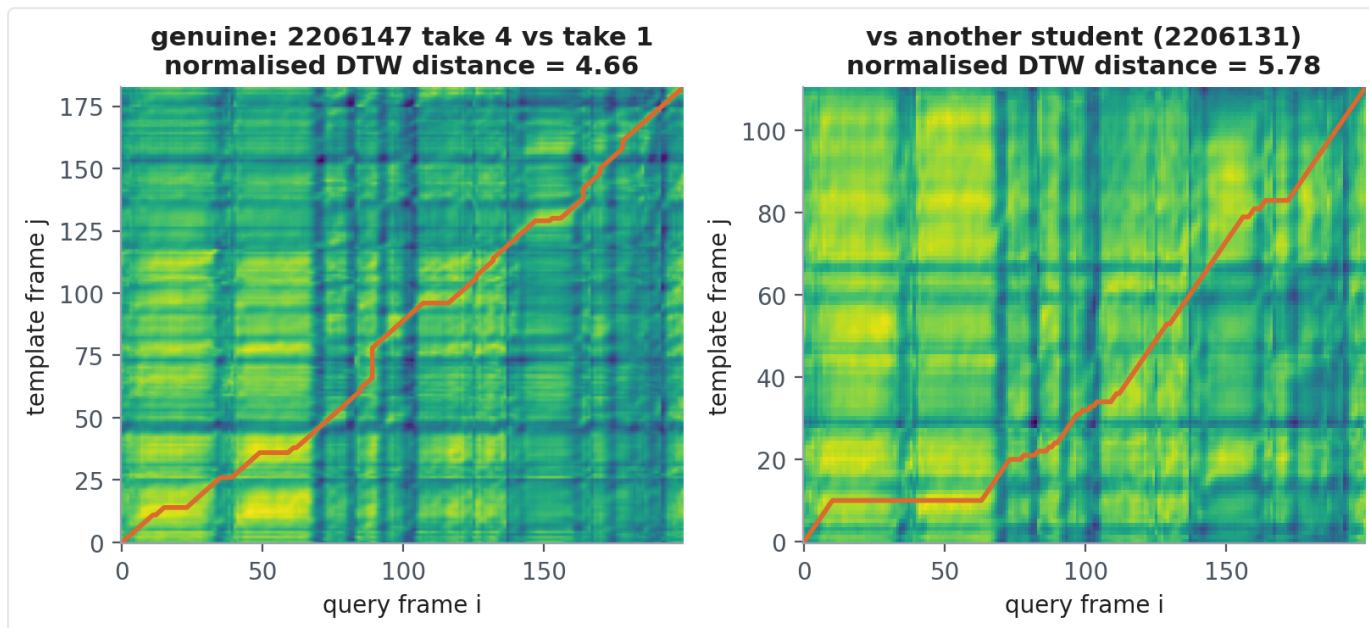


Figure 9. Local-cost matrices and optimal paths. The genuine pair (left) follows a low-cost valley near the diagonal (distance 4.66). Against another student's name (right) the path is forced into long flat detours (5.78).

4.2 Multi-template matching and cohort normalisation

$d_k = \min$ over all usable takes of student k of $\text{DTW}(\text{query}, \text{take})$ (v4.1.4: only take 1.wav)

$P(k) = \ln [\text{mean of the 5 smallest } d_j, j \neq k / d_k]$

Why the ratio works: a new microphone, a louder room or a cold multiplies *all* distances by roughly the same factor γ . Then $P(k) = \ln(\gamma \cdot \text{rivals} / \gamma \cdot d_k) = \ln(\text{rivals} / d_k)$ does not change. A fixed threshold on raw d_k (v4.1.4 used 31.35 and 30.35) fails as soon as $\gamma > 1$.

```
function P = cohort_lower(d, k) % in vsd_score_query.m
for c = 1:numel(d)
    o = d([1:c-1, c+1:end]); o = sort(o(isfinite(o)));
    P(c) = log(mean(o(1:min(k,numel(o))))) / max(d(c), 1e-9));
end
```

In plain words

$P_{\text{name}} = +0.17$ means "your name recording is $e^{0.17} \approx 1.19$ times closer to your own takes than to the average of your five closest rivals". Negative means someone else is closer.

What would happen if...

Band 0.1 instead of 0.4: a student who says his name 40 % slower than at enrolment cannot be aligned, and the distance explodes. **No band:** pathological paths (one query frame matched to half the template) become legal and imposters get cheaper matches. **Cohort size 1** (ratio to the single best rival): very noisy, because one lucky rival decides. **Cohort size 31** (all students): the normaliser is dominated by distant students and becomes insensitive. **Mean over templates instead of min:** an old take from a different mic pulls the genuine score down.

5. Who is speaking: GMM-UBM voice biometrics (vsd_gmm.m)

5.1 Gaussian mixture model

$p(x | \lambda) = \sum_{k=1}^K w_k \cdot N(x; \mu_k, \text{diag } \sigma_k^2)$, $K = 64$, $x = 24$ -D speaker feature frame

$\ln [w_k N(x; \mu_k, \sigma_k^2)] = \ln w_k - \frac{1}{2} \sum_d \ln(2\pi\sigma_{kd}^2) - \frac{1}{2} \sum_d (x_d - \mu_{kd})^2 / \sigma_{kd}^2$

$\ln p(x|\lambda)$ computed stably with the log-sum-exp trick: $m + \ln \sum_k \exp(L_k - m)$, $m = \max_k L_k$

Each Gaussian roughly models one class of sounds (a vowel region, a fricative, a nasal) *as this speaker produces it*. The mixture as a whole describes the speaker's "sound space", independent of what was said.

5.2 Universal Background Model trained by EM

E-step: $\gamma_t(k) = w_k N(x_t; \mu_k, \sigma_k^2) / \sum_j w_j N(x_t; \mu_j, \sigma_j^2)$ (responsibility of component k for frame t)

M-step: $n_k = \sum_t \gamma_t(k)$, $w_k = n_k / T$, $\mu_k = \sum \gamma_t(k) x_t / n_k$, $\sigma_k^2 = \sum \gamma_t(k) x_t^2 / n_k - \mu_k^2 + 10^{-3}$

The UBM is trained on the **isolated digits of all students** (up to 80 000 frames). This is text-independent speech that never contains a test phrase. Initialisation is deterministic: K frames spread evenly over the data, then 10 k-means iterations, then 30 EM iterations. So MATLAB and Octave give the same model every time.

5.3 MAP adaptation: one model per student from 10–20 s of speech

$n_k = \sum_t \gamma_t(k)$, $E_k[x] = \sum_t \gamma_t(k) x_t / n_k$ (student's frames, UBM responsibilities)

$\alpha_k = n_k / (n_k + r)$, $r = 16$ $\hat{\mu}_k = \alpha_k E_k[x] + (1 - \alpha_k) \mu_k^{\text{UBM}}$ (weights and variances shared with the UBM)

```
function mu = map_adapt(X, ubm, r)
L = log_joint(X, ubm.w, ubm.mu, ubm.var); % T x K log(w_k N(x_t))
m = max(L, [], 2);
P = exp(L - m); P = P ./ sum(P, 2); % responsibilities gamma_t(k)
n = sum(P, 1).'; % soft counts n_k
Ex = (P.' * X) ./ max(n, 1e-10); % E_k[x]
a = n ./ (n + r); % data-dependent adaptation factor
mu = a .* Ex + (1 - a) .* ubm.mu;
```

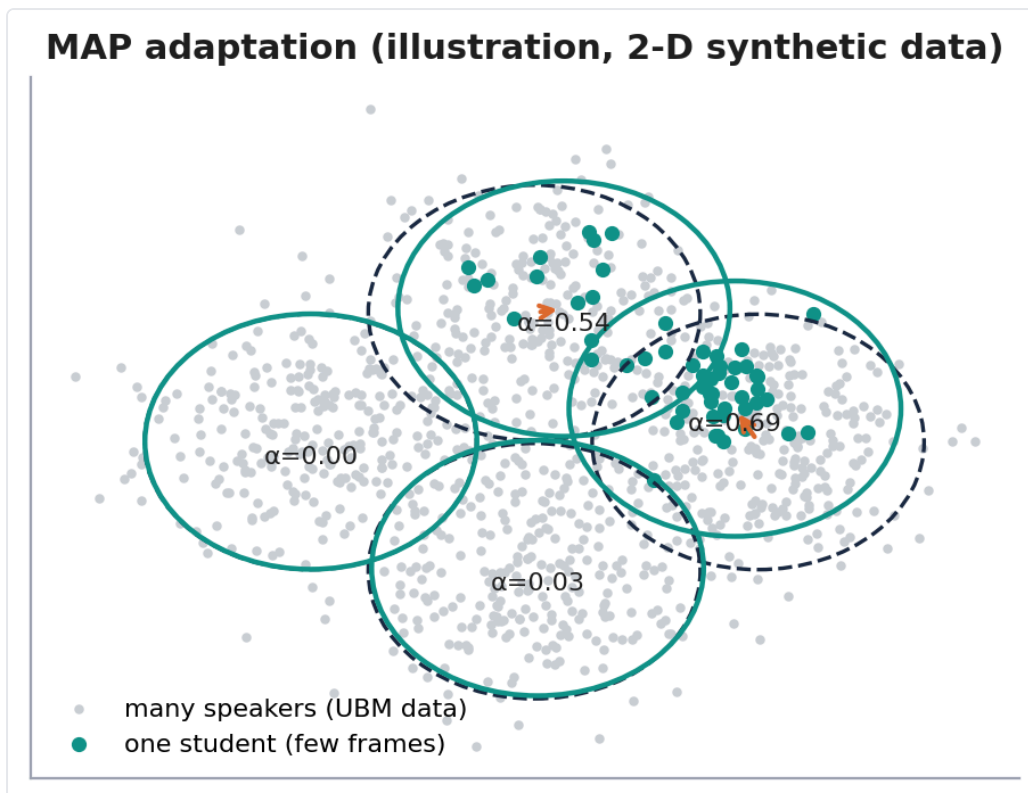


Figure 10. Illustration with synthetic 2-D data. Components that explain many of the student's frames move a long way ($\alpha \approx 0.7$). Components the student never used stay at the UBM mean ($\alpha \approx 0$), so a few seconds of speech cannot over-fit.

5.4 The voice score

$$\text{LLR}(k) = (1/T) \sum_t [\ln p(x_t | \lambda_k) - \ln p(x_t | \text{UBM})] \text{ (frames of the ID and the name together)}$$

$$V(k) = \text{LLR}(k) - \text{mean of the 5 largest LLR}(j), j \neq k \text{ (cohort normalisation, like P)}$$

In plain words

The UBM is "an average student on these microphones". The LLR asks: "does this sound more like student k than like an average student?" Subtracting the best rivals then asks: "...and more like k than like anyone else enrolled?" $V > 0$ for student k means k is the best voice.

What would happen if...

$K = 8$ (as in v4.1.4): each Gaussian must cover several sound classes, so speakers blur together. **$K = 32$** : measured cross-mic acceptance 65.2 % ($K = 64$: 78.3 %). **$K = 128$** : the same accuracy as 64 but twice the cost. **$r = 0$** : pure maximum-likelihood means from 10 s of data, which over-fit and are unstable. **$r \rightarrow \infty$** : every student equals the UBM, so all LLRs are 0. **The UBM trained on the ID/name phrases**: the test phrase is inside the "background", so LLRs shrink. **Adding the digits to each student's model**: tried; no gain (97.4 % LOTO either way).

6. The decision: fused score, imposter identification, unknown speaker (vsd_decide.m)

6.1 The rule

$F(k) = 0.5 \cdot P_{id}(k) + 1.0 \cdot P_{name}(k) + 0.25 \cdot V(k)$ candidate $c = \operatorname{argmax}_k F(k)$ (or the typed ID)

G1 quality: both phrases ≥ 0.35 s of speech and level $\geq 10^{-3}$

G2 phrase: $P_{name}(c) \geq 0.07$ and $P_{id}(c) \geq -0.15$

G3 voice: $V(c) \geq 0$ and c is the rank-1 voice

G4 imposter: $X = \operatorname{argmax}(P_{id} + P_{name})$ (words), $Y = \operatorname{argmax} V$ (voice). IMPOSTER if $Y \neq X$, $V(Y) - V(X) \geq 0.15$, $V(Y) \geq 0.20 \rightarrow$ report Y

CLEAR WINNER (v1.1): also VERIFIED if $F(c) - F(\text{rank } 2) \geq 0.08$, $P_{name}(c) \geq 0.12$, voice rank of $c \leq 3$, $V(c) \geq -0.10$, $P_{id}(c) \geq -0.15$

UNKNOWN if $\max V < 0$ and $\max P_{name} < 0.07$; otherwise RETRY with the failing gate named

```
F = c.WeightID*Pid + c.WeightName*Pn + c.Lambda*V;
[~, order] = sort(F, 'descend'); cidx = order(1);
[~, vorder] = sort(V, 'descend'); vrank = find(vorder == cidx, 1);
[~, X] = max(Pid + Pn); % who the WORDS name
Y = vorder(1); % whose VOICE it is
R.Gates.Phrase = (Pn(cidx) >= c.MinPname) && (Pid(cidx) >= c.MinPid);
R.Gates.Voice = (V(cidx) >= c.MinVoice) && (vrank <= c.VoiceRankMax);
isImp = (Y ~= X) && (V(Y) - V(X) >= c.ImpGap) && (V(Y) >= c.ImpMinVoice);
cw = Pn(cidx) >= c.ClearWin.MinPname && R.FMargin >= c.ClearWin.MinFMargin && ...
    vrank <= c.ClearWin.VoiceRankMax && V(cidx) >= c.ClearWin.MinVoice; % v1.1
if isImp
    R.Decision = 'IMPOSTER'; R.ImposterSuspect = st{Y};
elseif (R.Gates.Phrase && R.Gates.Voice) || cw
    R.Decision = 'VERIFIED'; % R.Rule = 'standard' or 'clear-winner'
elseif max(V) < c.MinVoice && max(Pn) < c.MinPname
    R.Decision = 'UNKNOWN';
else
    R.Decision = 'RETRY'; % Stage 'phrase' or 'voice' tells the user what to fix
end
```

6.2 Why these weights and thresholds

- **The ID phrase gets half weight and only a veto (-0.15 ; it was -0.10 in v1.0).** All roll numbers start with "2206...", so their DTW distances are nearly equal (see the score distributions below). A positive requirement on P_{id} would reject genuine students at random.
- **The name carries the word evidence** (threshold 0.07). Names differ a lot between students.
- **The voice must be rank 1**, unless the candidate is a clear winner (Section 6.5). This is what stops a friend who says your roll number and name perfectly.
- **$\lambda = 0.25$ for V in F .** V is on a larger numerical scale (genuine median about 1.3) than P (about 0.4). The weight keeps the words in charge of *choosing* the candidate, while the gates make both kinds of evidence *necessary*.
- The thresholds were selected on the corpus under the constraint "0 unknown speakers accepted", maximising genuine acceptance (LOTO + cross-mic) (`proto/tune2.py`). They were then frozen and checked in MATLAB/Octave.

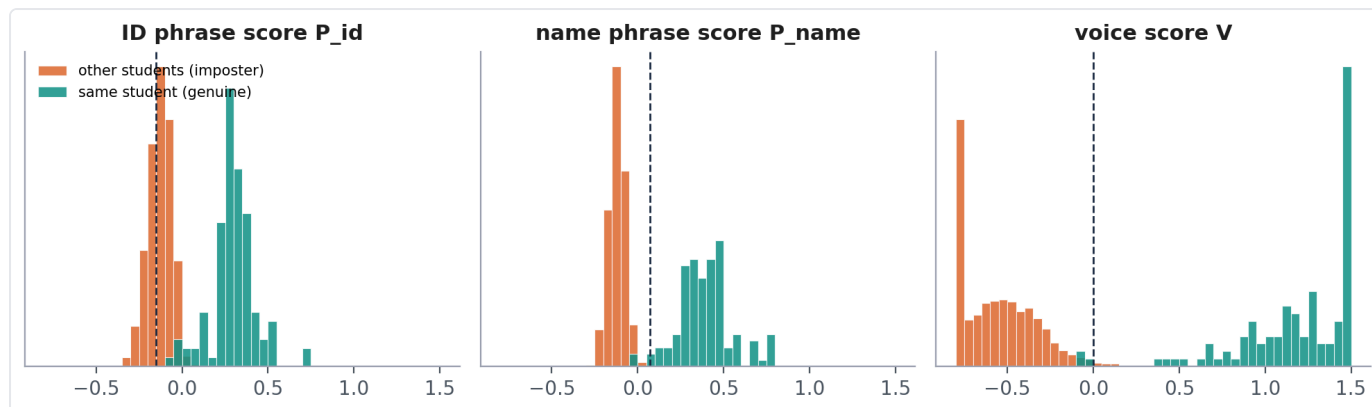


Figure 11. Genuine (same student) vs imposter (other students) scores over all leave-one-take-out trials. Name and voice separate cleanly. The roll-number score overlaps, which is why it only gets a veto. The dashed lines are the thresholds.

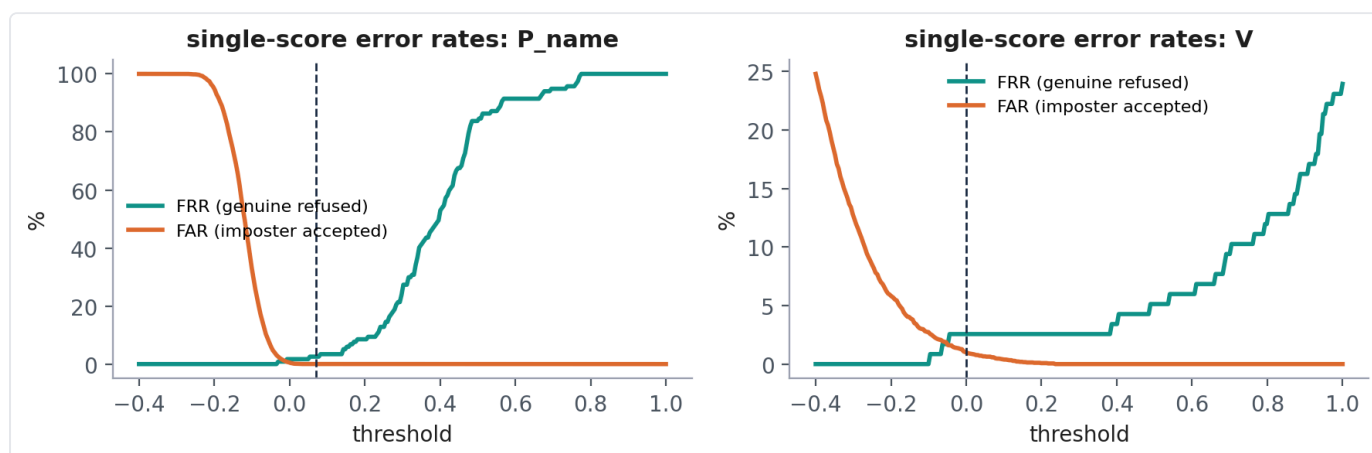


Figure 12. FAR and FRR of each single score as its threshold moves. Each score alone has an operating point with a few percent error; the gates combine them (AND), which pushes the imposter acceptance to 0 on the corpus while keeping genuine acceptance at 97 %.

6.3 Worked example 1: the student v4.1.4 kept rejecting (2206147)

Take 4 of 2206147 was recorded on our laptop microphone. It was held out of the enrolment and scored against all 32 students:

Score	2206147	Best other student	Threshold	Gate
P_id	+0.052 (DTW 4.79)	+0.033 (2206143, DTW 4.86)	≥ -0.15	passed
P_name	+0.167 (DTW 4.66)	-0.012 (2206137, DTW 5.42)	≥ 0.07	passed
V (voice)	+0.259 (rank 1)	+0.178 (2206138)	≥ 0 , rank 1	passed
Words X / voice Y	X = 2206147, Y = 2206147			no conflict

Result: **VERIFIED**, then the random-digit challenge "six six zero" was answered correctly (distance 4.19, margin 1.05), in the same voice. In v4.1.4 the same take had an ID margin of about 1.02 (limit 1.20) and was refused at the first gate.

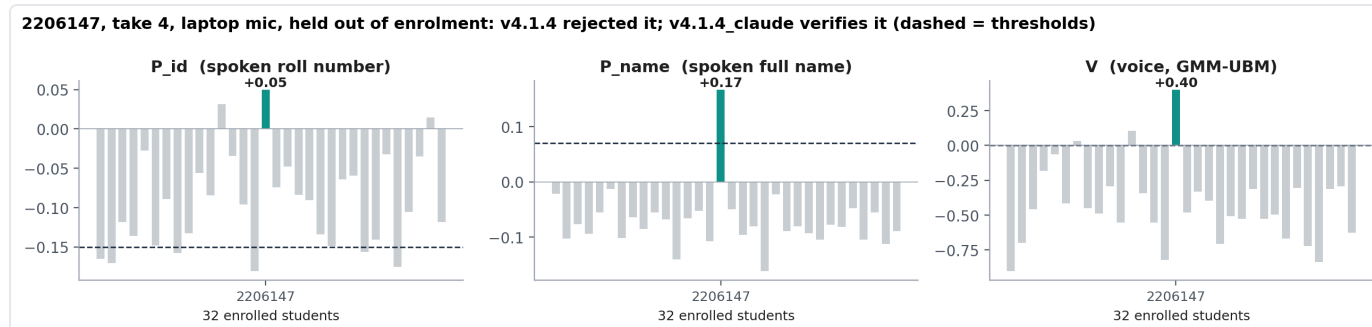


Figure 13. The three score vectors for this take: the roll number barely singles out 2206147 (+0.05), while the name (+0.17) and the voice (+0.40, Python mirror) clearly do.

6.4 Worked example 2: an imposter is named

2206147 says 2206141's roll number in his own voice (built from his own digits) and plays a recording of 2206141's name. Real output of `RUN_ME('demo')`:

```
Candidates (fused score = 0.5*P_id + P_name + 0.25*V):
#1 2206141 P_id +0.010 P_name +22.458 voice +0.227
#2 2206147 P_id +0.187 P_name -0.263 voice +0.757
* IMPOSTER ALERT: the spoken ID/name claim student 2206141, but the VOICE belongs
  to enrolled student 2206147 (voice score 0.76 vs 0.23). Access refused.
=> IMPOSTER 2206141 (likely imposter 2206147)
```

In plain words

The words say "2206141". The voice says "2206147", with a gap of 0.53, more than the 0.15 required. The system therefore does not simply refuse: it tells the hall office who was probably speaking. ($P_name = +22$ is so large because the name was a digital copy of a stored file. Had the phrases been accepted, the replay memory would have caught it next.)

What would happen if...

ImpGap = 0: any genuine student whose voice happens to be ranked 2nd would be labelled an imposter. That is unfair and would cause queue arguments. **ImpGap = 0.5:** a friend with a similar voice could pass the imposter check (they would still fail the voice gate G3). **No voice gate** (words only): measured unknown-speaker acceptance rises from 0 % to 6.0 %. **No word gate** (voice only): 100 % of unknown speakers are accepted, because *some* enrolled voice is always the closest. Cohort-normalised scores are relative, so the words are needed to reject strangers.

6.5 v1.1: the "clear winner" rule, and why we did not just lower the threshold

At the counter, a genuine student (2206141) was refused twice by the old v4.1.4 matcher even though the name phrase ranked him first both times. The log showed:

v4.1.4 log	Try 1	Try 2
ID phrase best match	2206157	2206144
Name phrase best match	2206141	2206141
Name lead (runner-up ÷ best distance)	1.209	1.194
Required fixed margin	1.212	1.212
Joint margin for 2206141	1.13×	1.09×
Decision	REFUSED: "ID and Name acoustic matches disagree"	REFUSED

Two things went wrong. The ID phrase is a near-tie for every student (all roll numbers start with 2206), so its top-1 was a random other student and it vetoed a correct name. And the name, which did point to the right student, missed a *fixed* margin by only 0.3 % and 1.5 %. The same pattern refused many students.

There were two ways to respond, and both were measured on the same trials (Python mirror of the engine, enrolment from take 1 only, 85 genuine / 117 unknown-speaker / 3627 perfect-words imposter trials):

Option	Genuine accepted	Unknown speaker accepted	Perfect-words imposter accepted
v1.0 rules	91.8 %	0.9 %	0.08 %
A. Lower the name threshold 0.07 → 0.05	91.8 %	3.4 % (2.6 % with all takes enrolled)	0.08 %
B. Relax the ID veto -0.10 → -0.15	92.9 %	0.9 %	0.08 %
C. Clear-winner ranking rule	94.1 %	0.9 %	0.25 %
v1.1 deployed = B + C	95.3 %	0.9 %	0.25 %

In plain words

Lowering a threshold helps everybody who is close to it, including strangers: option A accepted no extra genuine student but let 2.6–3.4 % of unknown speakers in. A clear winner is different: it asks whether rank 1 is **far ahead** of rank 2 on the fused score, with a strong name match and a voice that is at least in the top three. A stranger never produces a large lead for one enrolled student, so the stranger rate does not change.

VERIFIED (clear winner) $\Leftrightarrow F(c) - F(\text{rank } 2) \geq 0.08 \wedge P_{\text{name}}(c) \geq 0.12 \wedge \text{voice rank}(c) \leq 3 \wedge V(c) \geq -0.10 \wedge P_{\text{id}}(c) \geq -0.15 \wedge \neg \text{IMPOSTER}$

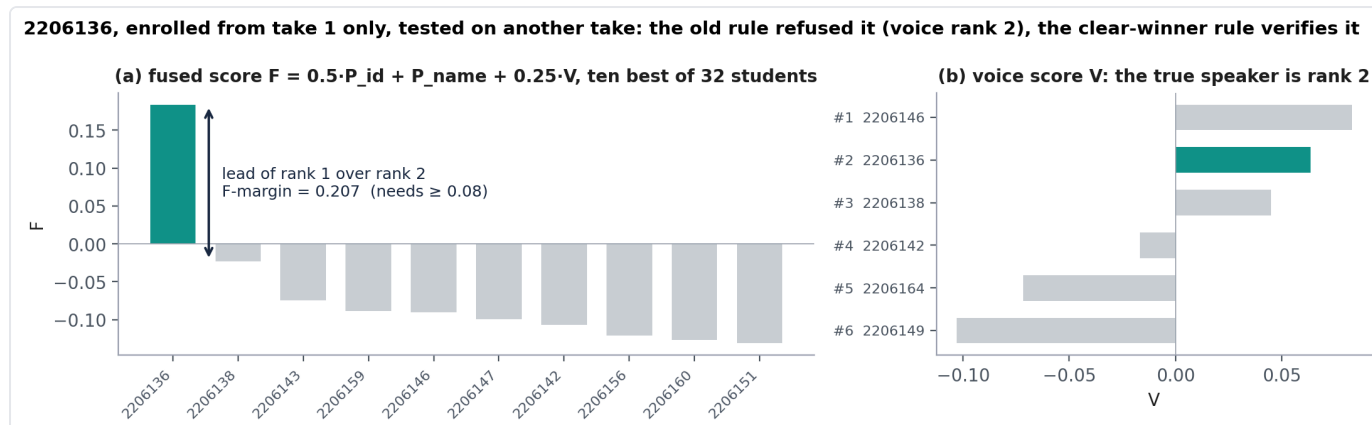


Figure 14. A real case from the corpus: 2206136, enrolled from take 1 only. The fused score puts him far ahead of everybody (lead 0.207, four times the 0.08 needed), but his voice is only rank 2, so the v1.0 voice gate refused him. The clear-winner rule verifies him; the imposter check still runs first.

The old v4.1.4 matcher (still used when `params.vsd.Enable = false`) received the same idea as a relative rescue in `speaker_verification_decision.m`: accept when the name ranks the candidate first with a lead ≥ 1.12 over the runner-up (and ≥ 1.15 over the mean of the 5 nearest), the joint margin is ≥ 1.08 , the candidate's ID distance is within 12 % of the best ID distance, the fused normalised distance is ≤ 1.60 (the ordinary ceiling is 1.15), and the voice ranks him in the top three. Both refused tries above clear the lead and joint-margin tests (1.209 and $1.194 \geq 1.12$; 1.13 and $1.09 \geq 1.08$).

What would happen if...

Lead 0.04 instead of 0.08: more near-ties are accepted, and imposter acceptance rises further. **Voice rank ≤ 5 :** a friend with a similar voice could ride on a correct name. **No P_{name} floor:** a clear lead built only from the roll number (which is nearly identical for everyone) would be accepted. The cost we accepted: the perfect-words imposter rate rose from 0.08 % to 0.22 % in MATLAB/Octave (0.25 % in the Python mirror).

6.6 Field log: an imposter the old matcher let in, and how v4.1.4_claude stops it

In a counter test, 2206147 said 2206150's roll number and name in his own voice. The old v4.1.4 matcher (the `Final project v4.1.4` folder) served him breakfast as 2206150:

Old matcher log	Value
Joint rank 1 / rank 2	2206150 1.13 / 2206163 1.21 (lead only 1.07×)
ID phrase best match	2206158 (2206150 only rank 3)
ID / name distance vs fixed limit	32.15 / 31.35 and 35.56 / 30.35, both above
Decision	VERIFIED 2206150: a meal served to an imposter

That matcher compares only the **words**. Anyone who says the right roll number and name looks like the owner to it. The voice model it had (8 Gaussians trained on one take) was too weak to object.

The v4.1.4_claude engine scores the **voice** as well. Each of 2206147's four held-out takes was scored as if its words were a perfect match for 2206150 (the mimic protocol of Section 11):

2206147 take	V(2206150)	Best voice	Decision
1 (database mic)	-0.82 (rank 28)	2206147, +1.36	IMPOSTER → 2206147
2 (database mic)	-0.85 (rank 28)	2206147, +1.40	IMPOSTER → 2206147
3 (database mic)	-0.84 (rank 28)	2206147, +1.47	IMPOSTER → 2206147
4 (laptop mic)	-0.46 (rank 16)	2206147, +0.26	IMPOSTER → 2206147

The voice gap $V(Y) - V(X)$ is at least 0.72, far above the 0.15 needed. Even when $V(Y)$ falls below 0.20 and the IMPOSTER label is not printed, the voice gate ($\text{rank} \leq 3, V \geq -0.10$) still refuses the claim.

The legacy matcher inside v4.1.4_claude (used only when `params.vsd.Enable = false`) now has the same protection: `capture_voice_features.m` keeps the raw recording, and `vsd_voice_veto.m` scores it with the GMM-UBM models before any meal is served. Tested on the stored files, it refused 2206147 → 2206150 and 2206141 → 2206150 on all four takes each and named the speaker. It passed every genuine take of 2206150 and 2206147.

In plain words

How to tell which engine you are running: every transaction in v4.1.4_claude starts with the line Engine :
v4.1.4_claude-1.1 Lines such as "Multi-modal joint biometric ranking" and "39-dimensional MFCC features" without an Engine line come from the old folder.

7. Replay-attack prevention: three layers

Threat model. An attacker wants a meal in another student's name. The attacker can:

- (a) speak the student's ID and name themselves (the imposter case of Chapter 6);
- (b) play a **recording** of the student from a phone;
- (c) inject a stored enrolment file digitally;
- (d) cut and paste the student's recorded digits to answer a challenge.

A replay of the real student has the *right words and the right voice*, so the identity gates alone cannot stop it. Liveness needs something a recording cannot know in advance.

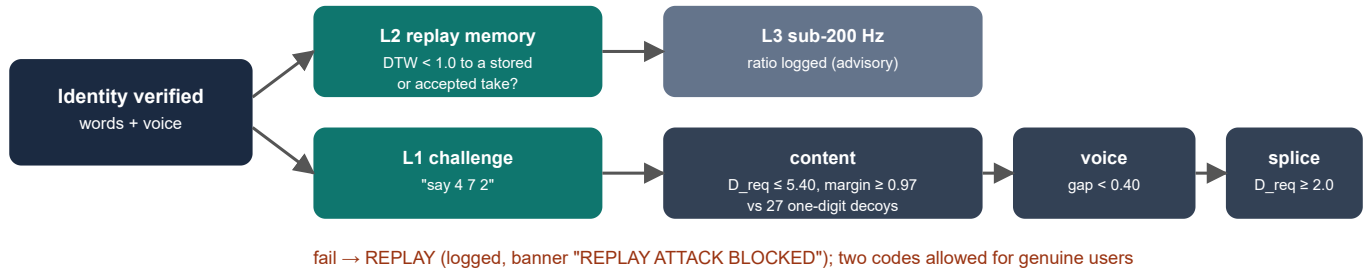


Figure 15. the three replay-prevention layers after identity verification.

7.1 Layer 1: the random-digit challenge (vsd_challenge.m)

After the identity is verified, the kiosk shows and speaks three random digits, for example "four seven two". The answer is checked in three ways.

Reference: $T(\text{code})$ = the claimed student's own isolated digit recordings joined in that order (0.2 s lead and tail, 0.05 s gaps, each at unit peak)

Decoys: the 27 codes that differ from the asked code in exactly one position (3 positions $\times$ 9 other digits), which are the hardest confusions

$D_{\text{req}} = \text{DTW}(\text{answer}, T(\text{code}))$, $D_j = \text{DTW}(\text{answer}, T(\text{decoy}_j))$

Content pass: $D_{\text{req}} \leq 5.40$ and $\text{margin} = \min_j D_j / D_{\text{req}} \geq 0.97$

Voice pass: $\text{gap} = \max_j V_{\text{ch}}(j) - V_{\text{ch}}(\text{claimed}) < 0.40$ (GMM-UBM on the digit frames)

Splice guard: $D_{\text{req}} < 2.00 \rightarrow$ the answer is a paste-up of the stored digit files $\rightarrow$ REPLAY

```

codes = [code; neighbours(code)];           % asked code + 27 decoys
for k = 1:size(codes,1)
    T = template(M.Digits{sid}, codes(k,:), c); % own digits concatenated
    D(k) = vsd_dtw(U.Content, T.Content, c.DtwBand);
end
info.Margin = min(D(2:end)) / max(D(1), 1e-9);
pass = (D(1) <= c.Challenge.MaxDist) && (info.Margin >= c.Challenge.MinMargin);
info.Splice = D(1) < c.Challenge.MinDist && strcmp(info.Mode, 'own');
if info.Splice, pass = false; end

```

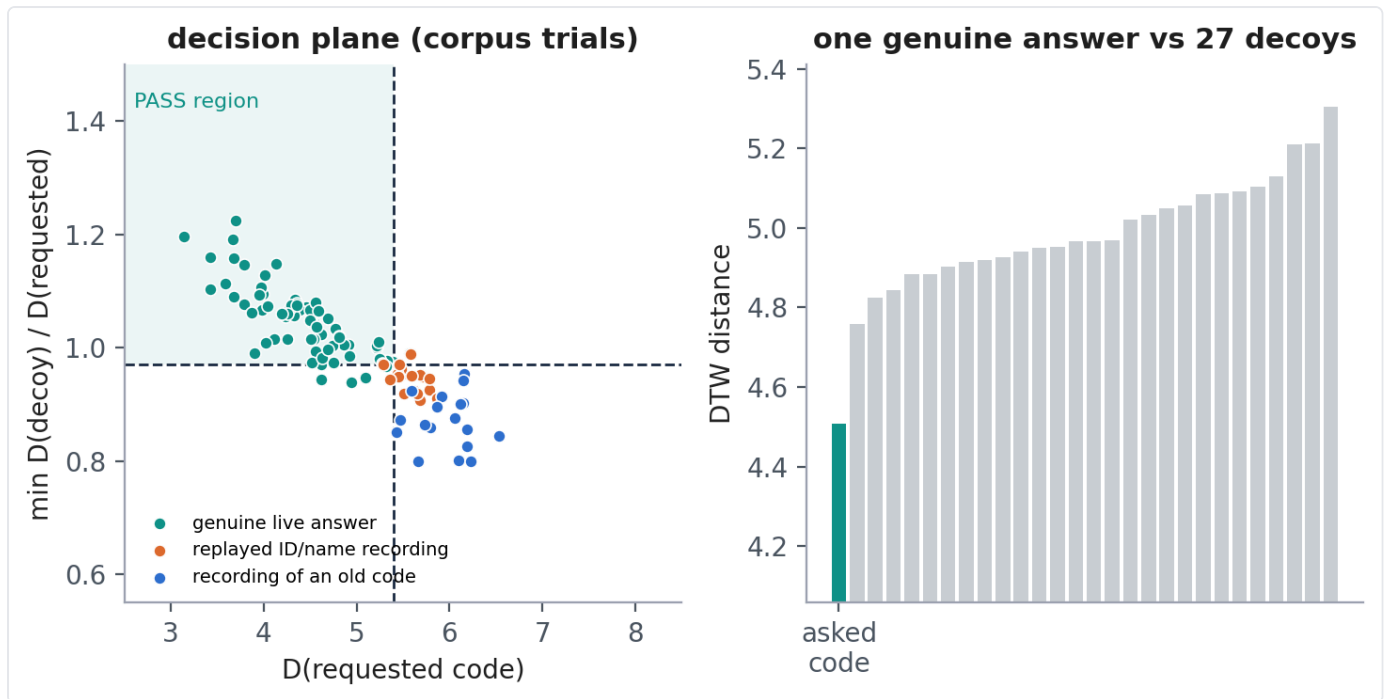


Figure 16. Left: every corpus trial in the (D_{req} , margin) plane. Genuine live answers (teal) fall in the PASS region; replayed ID/name recordings (orange) and recordings of an old code (blue) do not. Right: one genuine answer is closer to the asked code than to all 27 one-digit decoys.

Measured on the corpus	Result
Genuine answers, digit content (72 trials, connected counting takes vs isolated templates)	93 % pass per attempt (67/72)
Genuine answers, voice gap < 0.40 (54 trials, 18 speakers)	90.7 %; another student speaking the digits: 1.9 %
Replayed ID / name recordings offered as the answer	0 / 18
Recording of an older code	0 / 18
Smallest distance of a genuine live answer	3.58 (so the splice floor 2.0 never refuses a genuine answer)
End-to-end sweep (one transaction per student, two codes allowed)	16 of 17 testable students passed

In plain words

A recording made yesterday cannot contain "four seven two" if the kiosk chose those digits one second ago. Comparing against the 27 codes that differ by one digit makes the test strict: the answer must be *more like the asked code* than like every near-miss. Checking the voice of the digits stops a trick where the victim's recording plays the ID and name and the attacker then speaks the digits himself.

What would happen if...

One digit only: a 10 % chance that an old recording happens to contain it, and too short for a reliable voice check. **Six digits:** stronger, but the pass rate falls (each digit is another chance to fail) and the queue slows. **No decoys, only $D_{\text{req}} \leq \text{threshold}$:** any clear speech in the student's voice would pass, including an old code. **A fixed phrase instead of random digits:** it can be recorded once and replayed forever. **No splice guard:** an attacker with the enrolment files could paste "four", "seven", "two" together and pass with distance ≈ 0.01 . The demo's case 7 shows this being caught.

7.2 Layer 2: replay memory, the digital-copy detector (vsd_replay_guard.m)

flag if $\min \text{ over } \{ \text{all stored takes of the claimed student, last 20 accepted live takes} \} \text{ of } \text{DTW}(\text{query, take}) < 1.0$

Two natural repetitions are never identical. The smallest DTW distance between two genuine takes on the corpus was **2.41**, while an injected copy of a stored file scores **0.00**. Accepted utterances are remembered in `VSD_ReplayMemory.mat` (features only, no audio), so a recording of today's accepted transaction cannot be reused tomorrow.

7.3 Layer 3: the proposal's sub-200 Hz liveness heuristic (advisory)

$\text{ratio} = \Sigma_{20-200 \text{ Hz}} |X(f)|^2 / \Sigma_{20-3800 \text{ Hz}} |X(f)|^2$ on the native-rate recording; advisory flag if $\text{ratio} < 2.4 \cdot 10^{-4}$

Small phone loudspeakers cannot reproduce frequencies below about 200 Hz. The ratio is computed, logged and shown, but it does not refuse on its own. A microphone change moves it too, and there is no labelled corpus of real loudspeaker replays to calibrate it. Refusing students on an uncalibrated test would repeat v4.1.4's mistake.

Viva tip

"Why is your proposal's sub-200 Hz test only advisory?" Because a measurement you cannot calibrate must not make decisions. We replaced it with a test whose security rests on information, not on acoustics: a recording cannot know a random code.

8. Enrolment, the model cache and self-adaptation

8.1 Building the models (`vsd_build_models.m`, `vsd_models.m`)

1. List every WAV under `ID/` , `Name/` (including `Adaptive/`) and `Digits/` .
2. For each file, a cache key `path | bytes | date` is built. Only new or changed files are passed through `vsd_frontend` , and features are cached inside `VSD_Models.mat` .
3. A take is skipped if it is too quiet ($\text{RMS} < 10^{-3}$) or too short (< 0.35 s of speech). The build reports each skip, for example `Name/2206136/12.wav` (too quiet/short: rms 0.0027, speech 0.23s) .
4. The UBM is trained from the pooled digit frames and reused as long as the digit set and the configuration are unchanged.
5. One MAP-adapted model is built per student from all of their ID + name takes.
6. Before every transaction, `vsd_models` lists the folders (a few ms). If anything changed (Enrol tab, import, digit recorder, self-adaptation), the models update incrementally.

The first build on a new PC processes all 554 files (117 ID, 117 name, 320 digit recordings) in roughly one minute. Later starts load in about one second.

8.2 Self-adaptation to a new microphone (`vsd_adapt.m`)

Some transactions are verified with a large safety margin: $P_{\text{name}} \geq 0.12$, $V \geq 0.30$, and the challenge passed. Their ID and name recordings are stored in `ID/<id>/Adaptive/` and `Name/<id>/Adaptive/` . Three adaptive takes per phrase are kept, and older ones move to `Adaptive/archive` (never deleted, for audit). The next transaction therefore already knows how the student sounds *through the counter's current microphone*.

Watch out

Adaptation only happens after a passed challenge and with large margins. A replayed or imposter recording therefore cannot slowly "poison" a student's profile. Without these conditions, an attacker who once got lucky would be enrolled as the victim.

8.3 Recording the digits

Admin → Open DSP workspace → Enrol tab → type the Student ID → **Record digits 0-9** (`vsd_enrol_digits.m`). Each digit is recorded once with the same auto-stop capture and a quality check (≥ 0.15 s of speech), and saved as `Digits/<id>/<id>_<word>_NN.wav` . Students 2206135, 2206148 and 2206149 currently have no digits, so their challenge uses the digits of their three closest voices ("cohort" mode), which is weaker.

9. The complete transaction, the GUI and the dining rules

9.1 One counter transaction (vsd_verify_transaction.m + verify_meal_workflow.m)

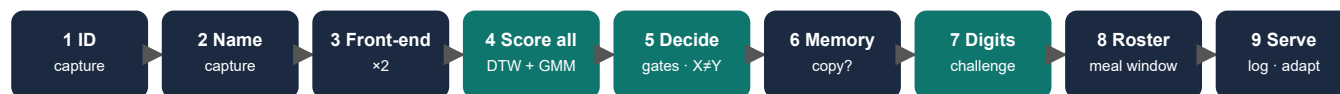


Figure 17. one counter transaction (vsd_verify_transaction + verify_meal_workflow).

1. **Capture ID** (`vsd_record_phrase`): 0.5 s silent ambient pre-roll, a spoken 3-2-1 cue, "speak now". Recording stops 0.8 s after speech ends (8 s cap). Instant Stop works throughout.
2. **Capture name** in the same way.
3. **Front-end** for both (Chapter 3). A phrase with less than 0.35 s of speech → "move closer and speak clearly" (stage quality).
4. **Score** the two phrases against every student (Chapters 4–5) and **decide** (Chapter 6). The GUI shows the top four candidates with P_id, P_name and V.
5. **Replay memory**: an exact copy → REPLAY.
6. **Challenge**: three random digits, up to two codes.
7. **Roster and meal** (`monthly_fee_entitlement` , `log_meal_csv`): assigned this month? inside a meal window? not yet served in this window?
8. **Serve**: welcome spoken aloud, `MealLog.csv` row, `VerificationAttempts.csv` row, replay memory updated, self-adaptation if confident.

9.2 What the operator sees: the six gates

Gate	Question	Typical refusal text
1 Signal quality	Are both recordings long and loud enough?	"The name recording is too quiet or too short (level 0.0008, speech 0.21 s)..."
2 Phrase content	Do ID + name single out one student?	"REFUSED (name +0.03)": say your name clearly and retry
3 Voice biometrics	Is the student the best-matching voice?	"REFUSED (voice -0.12, rank 3)"
4 Imposter check	Does the voice belong to the claimed student?	Banner IMPOSTER ALERT – voice matches 2206147
5 Anti-replay	Fresh digits, same voice, not a copy?	Banner REPLAY ATTACK BLOCKED – speak live
6 Roster and meal	Paid this month, meal window open, not served yet?	"NOT SERVED – 2206147: Meal time is over" / "Payment not done"

Other banners are **NOT ENROLLED – access refused** (UNKNOWN) and **VOICE NOT VERIFIED** (RETRY). The Enrol tab gains **Record digits 0-9** and **Rebuild voice models**. The Evaluation tab can run `vsd_evaluate_corpus` and `test_vsd_engine` , and shows their CSVs.

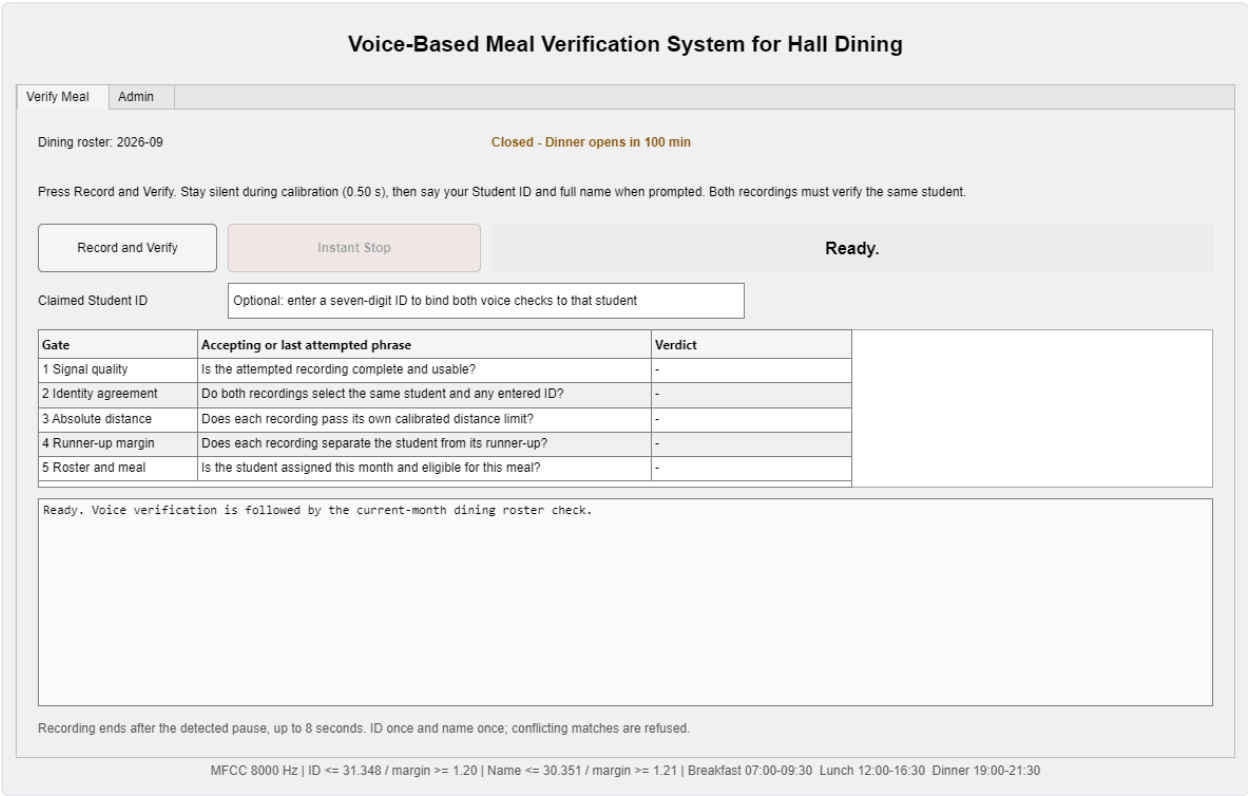


Figure 18. The counter window (the v4.1.4 shell, unchanged in layout). In v4.1.4_claude the gate table has the six rows above, and the status box prints the candidates, the challenge and the reason for any refusal.

9.3 Logs

File	One row per	Columns
MealLog.csv	served meal	StudentID, StudentName, Date, Time, Meal, Method, NameDistance, CouponDistance, Margin, IDDistance, IDMargin, NameMargin
VerificationAttempts.csv	attempt (accepted or refused)	Time, ClaimedID, VerifiedStudent, Decision, Verified, Served, Stage, Reason, IDDistance, NameDistance, IDMargin (= P_id), NameMargin (= P_name), IDTop5, NameTop5, Engine , VoiceScore , ImposterSuspect , ReplayFlag , Challenge (new)
VSD_ReplayMemory.mat	student	content features of the last 20 accepted utterances (no audio)

An old v4.1.4 attempts log with the previous header is renamed `VerificationAttempts_v4.1.4.csv` the first time the new version writes, so the two schemas never mix.

9.4 Dining business logic (kept from v4.1.4)

- **Monthly roster:** the admin assigns enrolled IDs for the current month (Admin tab, password protected). Only assigned students are served.
- **Meal windows:** `MealSchedule.csv` , editable in Admin → Meal hours (defaults 07:00–09:30, 12:00–14:30, 19:00–21:30, validated for overlap).
- **One meal per window:** `log_meal_csv` looks up today’s rows for the student and window (the anti-"double-dipping" rule of proposal modification 5).

10. Debugging v4.1.4: why genuine students were rejected, and what was changed

10.1 The evidence

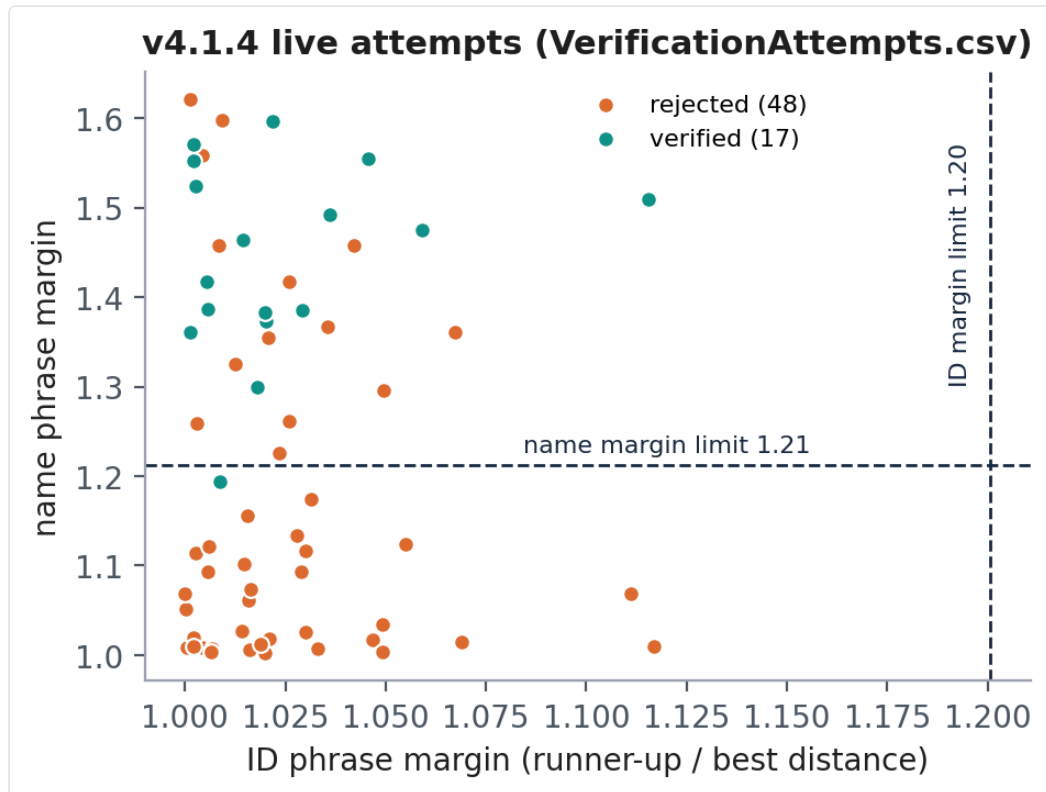


Figure 19. All 65 live attempts logged by v4.1.4 (VerificationAttempts.csv). Every attempt has an ID margin (runner-up distance / best distance) below the 1.20 limit, with a median of 1.02. The ID phrase almost never "won" clearly, so the rules that required it rejected 48 of 65 attempts, including genuine students such as 2206147.

10.2 Root causes found in the code

#	Where	What was wrong	Effect	Fix in v4.1.4_claude
1	<code>apply_voice_calibration.m</code>	Forced <code>params.enrollmentTemplateFileNames = {'1.wav'}</code>	One template per student, from one session and one mic	Allow-list removed; every usable take is a template, plus adaptive takes
2	<code>VoiceCalibration.mat</code>	<code>mfcc.CmsNormalise = 0</code>	Mic transfer function $H(\omega)$ left in every frame	CMVN always on in the new front-end
3	The data itself	Roll numbers 2206131...2206164 share 4–5 of 7 digits	ID-phrase DTW nearly equal for everyone (e.g. 4.79 vs 4.86)	ID gets half weight and a veto only; the name and voice carry the decision
4	<code>speaker_verification_decision.m</code>	"ID top-1 must equal name top-1", plus a margin ≥ 1.20 on each	A random ID winner vetoed a correct name	Fused score F with separate gates
5	Calibration	Fixed raw DTW ceilings 31.35 / 30.35	A new mic or a louder room shifts all distances, so everyone is rejected	Cohort ratios P and V are invariant to common shifts
6	<code>VoiceGMM_Model.mat</code>	8 mixtures, one take per student, diagnostic only	No real voice evidence; imposters undetectable	64-mixture GMM-UBM, MAP $r = 16$, used as gate 3 and for imposter naming
7	<code>dsp_parameters.m</code>	Error raised if the calibrated profile folders were missing	The whole GUI could not start after moving the folder	Warning only; the profile root is found automatically

10.3 File-by-file changelog

File	Status	Change
<code>vsd_config.m</code>	new	All engine settings with comments (front-end, DTW band, GMM, thresholds, challenge, replay, adaptation) and DataRoot auto-discovery
<code>vsd_frontend.m</code>	new	Resample → DC → spectral subtraction → normalise → pre-emphasis → MFCC → speech mask → Δ → CMVN
<code>vsd_dtw.m</code>	new	Anti-diagonal vectorised DTW with band and path-length normalisation
<code>vsd_gmm.m</code>	new	UBM (k-means + EM), MAP adaptation, LLR scoring
<code>vsd_build_models.m</code> , <code>vsd_models.m</code>	new	Incremental enrolment cache, UBM reuse, automatic update when files change
<code>vsd_score_query.m</code> , <code>vsd_decide.m</code>	new	Cohort scores; fused decision with the clear-winner rule (1.1); IMPOSTER / UNKNOWN / RETRY with reasons
<code>vsd_challenge.m</code> , <code>vsd_replay_guard.m</code>	new	Random digits with 27 decoys, splice guard; replay memory; sub-200 Hz advisory
<code>vsd_record_phrase.m</code> , <code>vsd_verify_transaction.m</code>	new	Raw-audio capture (same auto-stop policy) and the full transaction
<code>vsd_adapt.m</code> , <code>vsd_enrol_digits.m</code>	new	Self-adaptation store; digit recorder
<code>vsd_evaluate_corpus.m</code> , <code>vsd_transaction_sweep.m</code> , <code>vsd_calibrate_challenge_voice.m</code> , <code>test_vsd_engine.m</code>	new	Evaluation protocols, per-student end-to-end sweep, challenge-voice calibration, 7-case demo
<code>verify_meal_workflow.m</code>	changed	Runs the VSD engine and maps its result (Decision, Stage, ImposterSuspect, ReplayFlag, Challenge, VoiceScore) onto the existing result struct; the legacy matcher remains as a fallback
<code>dsp_parameters.m</code>	changed	<code>params.vsd = vsd_config(...)</code> ; all train folders point to the engine's DataRoot; calibration errors become warnings
<code>apply_voice_calibration.m</code>	changed	1.wav-only rule removed; missing folders only warn
<code>meal_verification_gui.m</code>	changed	Six-gate table, IMPOSTER/REPLAY/UNKNOWN banners and speech, digit recorder, model rebuild, evaluation entries
<code>log_verification_attempt.m</code>	changed	New columns; old-schema file archived
<code>vsd_voice_veto.m</code>	new (1.1)	GMM-UBM voice check applied to every accept of the legacy matcher; refuses and names an imposter
<code>speaker_verification_decision.m</code> , <code>capture_voice_features.m</code>	changed (1.1)	Relative clear-winner rescue for the legacy matcher; raw capture kept for the voice veto
<code>RUN_ME.m</code>	changed	Removes sibling project folders from the MATLAB path (no function shadowing); modes <code>evaluate</code> , <code>demo</code> , <code>rebuild</code>

Watch out

If you add both "Final project v4.1.4" and "..._claude" to the MATLAB path, functions with the same name shadow each other and you may silently run the old matcher. Always start with `RUN_ME` from inside the `_claude` folder; it cleans the path first.

11. Evaluation: protocols, outputs and how to read them

11.1 The data

There are 32 enrolled students (2206131–2206164), with 117 ID takes, 117 name takes and 320 digit recordings. The recordings come from three microphones, which the evaluation infers from the file format:

- **db**: the original database mic, 48 kHz stereo/float, 88 test pairs.
- **usb**: a second USB microphone, 48 kHz mono int16.
- **laptop**: our GUI laptop mic, 44.1 kHz.

Together, the usb and laptop microphones give 28 test pairs.

11.2 Protocols

Protocol	What is enrolled	What is tested	Measures
LOTO (leave-one-take-out)	Everything except the tested ID+name pair (its voice model is re-adapted without it)	Every take pair, all mics	Genuine acceptance; wrong-student accepts
Cross-mic	Only database-mic takes of every student	Pairs recorded entirely on other mics	The "mic changed" complaint
Unknown speaker	All students except one	The removed student saying their own ID and name	False acceptance of strangers
Perfect mimic	All	Every genuine utterance of A scored as if A said victim B's words perfectly (words gate forced to pass)	Voice-only security; imposter naming
Take-1 comparison	Take 1 only (as v4.1.4)	All other takes; unknown speakers	Fair comparison with the senior baseline and v4.1.4
Transaction sweep	All except the tested take	One full transaction per student, including a random challenge answered with digits cut from their counting recording	End-to-end behaviour
Demo (<code>RUN_ME('demo')</code>)	As deployed	7 scripted cases	Genuine, imposter and replay behaviour

11.3 Results

```
v4.1.4_claude corpus evaluation 2026-09-30 09:34:10 (engine 1.1)
LOTO genuine acceptance: 114/116 = 98.3 % (wrong-student accepts: 0)
  same (database) mic: 88/88, other mics: 26/28
CROSS-MIC (enrol database mic only): 18/20 = 90.0 % (wrong accepts 0)
UNKNOWN-SPEAKER false accepts: 0/116 = 0.0 %
PERFECT-MIMIC imposter trials: 3596, false accepts 0.22 %, imposter correctly named 97.4 %
```

Engine 1.0 (before the clear-winner rule and the -0.15 ID veto) gave 113/116 = 97.4 % LOTO and 0.08 % mimic acceptance on the same trials.

In plain words

This is Results/claude_eval/summary.txt, read line by line.

- **114/116**: in 116 genuine tests, each with the tested recording hidden from the system, the right student was accepted 114 times. It never accepted the wrong student.
- **18/20**: when the system only knows a student's voice from the database microphone and the student then speaks into a different microphone, 18 of 20 attempts are still accepted.
- **0/116**: none of the 116 attempts by a person the system has never enrolled were accepted.
- **0.22 %**: even if an imposter said someone else's roll number and name perfectly, the voice check let through only 8 of 3596. In 97.4 % of those attempts the system named the real speaker as the likely imposter.

Take-1 enrolment (fair comparison)	Same mic GAR	Other mic GAR	Unknown FAR
Senior baseline: VQ-LBG 8 codewords, 44.1 kHz, 12 MFCC+ Δ + $\Delta\Delta$	87.9 %	25.9 %	15.4 %
v4.1.4: DTW, fixed thresholds, CMS off	84.5 %	7.4 %	1.7 %
v4.1.4_claude 1.1	100 %	85.2 %	0.9 %

(v1.0, before the clear-winner rule: 100 % / 74.1 % / 0.9 %.)

The take-1 comparison uses 58 same-mic and 27 other-mic genuine trials and 117 unknown-speaker trials. It was run with the Python mirror, where the senior and v4.1.4 pipelines were re-implemented from their code.

The ablation below was run with the engine 1.0 decision rules (before the clear-winner rule).

Decision ablation (same features, same trials)	LOTO	Cross-mic	Unknown accepted
Words only: ID and name must agree (v4.1.4 rule)	95.7 %	65.2 %	13.7 %
Words only: fused P_id + P_name with gates	97.4 %	78.3 %	6.0 %
Voice only: GMM-UBM, $V \geq 0$	97.4 %	78.3 %	100 %
Agreement rule + voice gate	95.7 %	65.2 %	6.0 %
Deployed: fused words + voice + imposter check	97.4 %	78.3 %	0.0 %

(The Python ablation's cross-mic set has 23 trials, while the MATLAB set has 20. That is why the deployed row shows 78.3 % here and 90.0 % above.)

11.4 Demonstration output (RUN_ME('demo'))

```
>> RUN_ME('demo')
...
---- 3 IMPOSTER: 2206147 says 2206141's roll number ----
ID: 2.40 s of speech, 240 frames, level 0.1561.
Name: 1.31 s of speech, 131 frames, level 0.1715.
Scoring phrase content (DTW) and voice (GMM-UBM) against all enrolled students...
Candidates (fused score = 0.5*P_id + P_name + 0.25*V):
#1 2206141 P_id +0.010 P_name +22.458 voice +0.227
#2 2206147 P_id +0.187 P_name -0.263 voice +0.757
#3 2206136 P_id -0.059 P_name -0.239 voice -0.150
#4 2206158 P_id -0.114 P_name -0.186 voice -0.279
* IMPOSTER ALERT: the spoken ID/name claim student 2206141, but the VOICE belongs to enrolled student 2206147 (voice s
core 0.76 vs 0.23). Access refused.
=> IMPOSTER 2206141 (likely imposter 2206147)
...
===== v4.1.4_claude engine demonstration =====
[PASS] 1 GENUINE (own ID + name + live digits) -> VERIFIED 2206147
[PASS] 2 IMPOSTER: 2206147 types 2206141's ID -> IMPOSTER 2206141 (likely imposter 2206147)
[PASS] 3 IMPOSTER: 2206147 says 2206141's roll number -> IMPOSTER 2206141 (likely imposter 2206147)
[PASS] 4 REPLAY: 2206141's stored files injected -> REPLAY 2206141
[PASS] 5 REPLAY: 2206141 played through a phone speaker -> RETRY 2206141
[PASS] 6 STALE CHALLENGE: old code 2 1 9 replayed for 5 6 6 -> REPLAY 2206141
[PASS] 7 SPLICE: 2206141's enrolled digit files pasted as the answer -> REPLAY 2206141
7/7 cases behave as designed.
```

Figure 20. Real console output. Case 3 is shown in full, followed by the summary of all seven cases.

Case	What is attempted	Decision	Layer that acted
1	2206147 says his own ID and name (laptop-mic take held out) and answers the digits live	VERIFIED	all gates passed
2	2206147 types 2206141's ID and speaks	IMPOSTER → 2206147	gate 4
3	2206147 says 2206141's roll number and plays his name	IMPOSTER → 2206147	gate 4
4	2206141's stored enrolment files injected digitally	REPLAY	replay memory (distance 0.00)
5	2206141's recordings played through a simulated phone speaker (250 Hz–5 kHz, clipping, room echo, noise)	RETRY (refused)	gate 2: the distortion already spoils the name match. For another student (2206131) the phrases passed and the challenge refused it.
6	A recording of an old code "2 1 9" offered for "5 6 6"	REPLAY	challenge content
7	2206141's enrolled digit files pasted together as the answer	REPLAY	splice guard (distance 0.01 < 2.0)

11.5 Transaction sweep: one full transaction per student

Result: 30 of 32 students were verified end-to-end. Of the 17 students whose counting recording could be cut into ten digits, 16 passed the challenge within two codes.

- **2206145** failed both codes, which the system reported as REPLAY. A genuine student would simply press Record again.
- **2206150's** take 4 name score was 0.05, below the 0.07 threshold, so the result was RETRY.

The full table is in `Results/claude_eval/transaction_sweep.csv`.

11.6 Output files

File (Results/claude_eval)	Columns and how to read them
loto_trials.csv , cross_mic_trials.csv	student, take, mic, decision, accepted, correct, wrong_accept, P_id, P_name, V, voice_rank, imposter_suspect. A row with correct = 0 is a false rejection: look at which score is below its threshold.
unknown_speaker_trials.csv	The same columns, but with P and V taken as the <i>maximum</i> over the remaining students. decision should be UNKNOWN or RETRY.
transaction_sweep.csv	student, take, decision, stage, verified, challenge (code and outcome), P_id, P_name, V
challenge_voice_calibration.csv	claimed, speaker, genuine, voice_gap, D_requested. Source of the 0.40 voice-gap and 2.0 splice thresholds.
summary.txt	The five headline lines above

12. A to Z: every improvement, and why

Area	Senior baseline	v4.1.4	v4.1.4_claude	Why it matters
Acquisition	Fixed 3 s, 44.1 kHz	0.5 s ambient, 3-2-1 cue, auto-stop 0.8 s, 8 s cap	Same (vsd_record_phrase), raw audio kept	Short names are not padded with 2 s of noise; long names are not cut
Sampling rate	44.1 kHz	8 kHz (rational)	8 kHz, any input rate (44.1, 48, 16 kHz)	5.5× less computation; the band the features use
Level	None	AGC (RMS)	Peak normalisation + CMVN	Distance to the mic does not change the features
Noise	None	Spectral subtraction with a pre-roll	Spectral subtraction, noise from the quietest 10 % of frames	Works on stored files and early speakers too
Endpointing	4096-sample block energy	Hybrid STE + ZCR	30 dB log-energy speech mask on frames	Frame-exact; keeps soft consonants
Pre-emphasis	0.99	0.97	0.97	Balanced tilt
Features	12 MFCC + Δ + $\Delta\Delta$, 50–7000 Hz	13 MFCC + Δ + $\Delta\Delta$, 300–3700 Hz, CMS off	c1–c12 + Δ , 100–3800 Hz, CMVN	The microphone cancels (Chapter 3.6)
Templates	1 VQ codebook	1.wav only	All takes + adaptive takes	Covers sessions and microphones
Word matcher	VQ distortion	DTW / (n+m), raw thresholds	DTW / path length, cohort score	Thresholds survive a channel change
Voice model	None	8-mix GMM, diagnostic only	64-mix GMM-UBM, MAP	Stops a friend who knows your words
Decision	Nearest codebooks agree	Both phrases pass their own gates, ID top-1 = name top-1	Fused F + separate gates	No random veto by the roll number
Unknown speaker	Accepted as the nearest student	Mostly refused	UNKNOWN decision	Non-residents cannot eat
Imposter	Not detected	Not detected	Named ("likely imposter = Y")	Proxy meals become visible to the hall office
Replay	None	Sub-200 Hz heuristic	Random digits + 27 decoys + voice + splice guard , replay memory, sub-200 Hz advisory	Security from information a recording cannot have
New microphone	Fails	Fails	CMVN + cohort + self-adaptation	The kiosk improves with use
Robustness of start-up	—	Error if folders moved	Profile root auto-discovered, warnings only	The kiosk always opens
Evaluation	Informal	Experiments on take 1	LOTO, cross-mic, unknown, mimic, sweep, demo, ablation	Every claim has a number behind it
Business logic	Gate attendance	Roster, meal windows, one meal per window, CSV	Unchanged + richer attempt log	Audit of every refusal

13. "What would happen if...": consolidated scenarios

Many examiners test understanding with counterfactuals. Each row gives the change, the predicted effect with its reason, and (where we measured it) the evidence.

If we...	...then (and why)
processed at 44.1 kHz instead of 8 kHz	The same accuracy (the mel bands stop at 3.8 kHz anyway), about 5.5× more samples to filter and frame. Frame lengths in samples change, so the FFT size must grow to 2048.
resampled with <code>x(1:6:end)</code> from 48 kHz	Aliasing: energy above 4 kHz folds into the speech band and fricatives smear. A low-pass before decimation is mandatory.
removed spectral subtraction	In a quiet room little changes. In the dining hall the noise fills the pauses, the speech mask keeps noise frames, and DTW distances rise for everyone. The cohort ratio partly compensates, but the name margin shrinks.
set $\alpha = 4$ or $\beta = 0$	$\alpha = 4$ deletes weak consonants. $\beta = 0$ produces musical noise (isolated tonal blips) that DTW matches as if it were speech.
removed pre-emphasis	F3/F4 are weighted 10–20 dB lower, speaker cues weaken and the voice gap between similar students shrinks.
used a rectangular window	–13 dB side lobes: strong low harmonics leak into the higher mel bands and the cepstra become noisier.
used 50 ms frames	The spectrum smears across phonemes (the vocal tract moves within 50 ms), and DTW loses temporal resolution. 10 ms frames: too few samples for a stable spectrum (a 100 Hz pitch period is 10 ms).
kept c0	Loudness enters the features, and standing 10 cm closer changes the match. This was the proposal's own complaint about the seniors.
used 20 cepstra	Measured: cross-mic acceptance fell from 78.3 % to 52.2 %, because high quefrequencies encode pitch harmonics and microphone ripple.
switched CMVN off (as v4.1.4 did)	The mic term h stays in the features and a new microphone shifts every distance. With CMS only on the voice features, measured cross-mic voice identification was 60.9 % (CMVN: up to 91.3 %).
dropped the deltas	Words with similar static spectra but different dynamics ("one" vs "nine") become more confusable in the challenge. Deltas are also offset-free.
kept only 1.wav per student	Exactly v4.1.4's failure: one session and one microphone. Take-1 other-mic acceptance: 85.2 % with the new engine (v1.1) vs 90–93 % with all takes.
used the mean DTW distance over templates instead of the minimum	An old take from another mic drags the genuine score down, and an extra take can then hurt.
set the DTW band to 0.1	Tempo changes of more than 10 % cannot be aligned, so slow or fast speakers are rejected.
removed the band	Degenerate paths let a short query match a small piece of a long template, so imposters get cheaper matches and the cost rises.
used raw DTW thresholds instead of cohort scores	The same brittleness as v4.1.4: any factor that scales all distances (mic, room, cold) moves every student across the threshold together.
simply lowered the name threshold 0.07 → 0.05	Measured: no extra genuine student accepted (take-1 enrolment 91.8 % either way), but 2.6 % of unknown speakers accepted instead of 0 %. The clear-winner rule was chosen instead (+3.5 points, strangers unchanged).
demand "ID top-1 = name top-1"	Measured: LOTO 95.7 % (vs 97.4 %), cross-mic 65.2 % (vs 78.3 %), unknown FAR 13.7 % (vs 0 %) with no voice gate.
used only the voice (no words)	Measured: 100 % of unknown speakers accepted. Cohort-normalised scores are relative, and some enrolled voice is always closest.

used only the words (no voice)	Measured: 6.0 % of unknown speakers accepted, and a friend who knows your roll number and name is accepted.
used 8 or 32 Gaussians	8: speakers blur together. 32: cross-mic 65.2 % measured. 64 is the smallest size that reached the best accuracy, and 128 adds cost with no gain.
trained the UBM on the ID and name phrases	The test words are then "background", so genuine LLRs shrink, and the UBM changes whenever a student enrolls.
set the MAP relevance to $r = 0$ or $r = 1000$	$r = 0$: over-fitted means from a few seconds of speech. $r = 1000$: every student is almost the UBM, so all LLRs are ≈ 0 .
set ImpGap = 0	Genuine students whose voice ranks 2nd by chance are called imposters, which is unfair and embarrassing.
removed the random-digit challenge	A phone recording of the student passes the identity gates. That is exactly what a replay is.
used a 1-digit or a 6-digit challenge	1 digit: a 10 % chance an old recording contains it, and too short for the voice check. 6 digits: more failures for genuine users and a slower queue.
removed the 27 decoys	Any clear digit string in the right voice would pass, including an old code.
removed the splice guard	The digit files themselves (for example leaked from the kiosk) could be pasted together into any code. Demo case 7 shows this.
made the sub-200 Hz test a hard gate	A different microphone or a small USB mic with a weak bass response would refuse genuine students. It is uncalibrated without a replay corpus.
adapted on every accepted transaction	A lucky imposter would become part of the victim's profile (template poisoning). This is why adaptation needs large margins and a passed challenge.
the student has a cold	Nasal resonances change. The words still match (cohort scores), and the voice score drops but usually stays rank 1. If refused, the student retries, and a confident accept adapts the profile.
the student whispers	No voicing: the voice features change completely and the speech mask keeps few frames. Refused with "speak louder". Whisper is out of scope.
two students have very similar voices (twins)	V gaps become small, so the imposter check may not fire. The words still differ, but a twin who knows the words may pass. This is a known limitation of voice biometrics.
an attacker uses a speech synthesiser trained on the victim	The synthesiser can say fresh digits. This "logical access" attack is out of scope and would need a learned countermeasure (ASVspoof-style CQCC/LFCC classifiers).
the hall has 500 residents	DTW against every take grows linearly: about 5000 alignments per phrase. The fix is to pre-select the top 20 by voice score (fast) and then run DTW on those only.

14. Limitations and future work

14.1 Limitations

- **Small corpus.** Only 28 genuine test pairs on non-database microphones, so the cross-mic rates carry $\pm 10\text{--}15\%$ uncertainty.
- **New microphones.** A student enrolled only on another microphone is accepted about 9 times in 10 at first. Self-adaptation and one enrolment take on the counter microphone close the gap.
- **Challenge cost.** It adds about 3 s per student, and genuine users fail about 1 code in 6 and must say a second code.
- **Missing digits.** 2206135, 2206148 and 2206149 have no digits, so their challenge is in cohort mode.
- **No real replay corpus.** The replay tests use digital copies and a simulated loudspeaker.
- **Out of scope.** Synthetic speech, voice conversion and skilled vocal mimicry.
- **Octave vs MATLAB.** The validation was run in GNU Octave 8.4 (identical code). The GUI's new banners and six-gate table were checked by code review; a live MATLAB session should record screenshots for the report.

14.2 Future work

- Record 1–2 takes per student on the counter microphone, and digits for every student.
- Collect real replays (phones, Bluetooth speakers) and train a countermeasure: CQCC/LFCC + GMM, or a small CNN.
- i-vector / x-vector embeddings with cosine scoring for large halls; voice-based pre-selection before DTW.
- Score normalisation per microphone (Z-norm / T-norm) and noise-adaptive thresholds.
- A Raspberry Pi kiosk with a USB microphone, and roster integration with the hall office's payment records.

15. Viva questions and answers

Answers are deliberately short, the way you would say them in the viva. Section numbers point to the full explanation.

A. Signals and sampling

Q1. What is the Nyquist rate and why is 8 kHz enough for this project?

A band-limited signal must be sampled at more than twice its highest frequency. Speech formants that carry the words and most of the speaker's identity lie below ~ 3.5 kHz, so $f_s = 8$ kHz (band 0–4 kHz) suffices; telephone speech uses the same rate.

Q2. What is aliasing and how does resample prevent it?

Components above $f_s/2$ fold back to lower frequencies after sampling. `resample(x, 80, 441)` up-samples by 80, applies a low-pass anti-alias FIR with cut-off $\pi/441$ (i.e. 4 kHz at the output), and keeps every 441st sample, so nothing above 4 kHz survives to fold.

Q3. Why is 44.1→8 kHz a *rational* resampling problem?

$8000/44100 = 80/441$ is not an integer ratio, so plain decimation is impossible; we need interpolation by $L = 80$ and decimation by $M = 441$ with one filter in between (polyphase form computes only the kept samples).

Q4. What does one FFT bin represent in our front-end?

$N_{\text{FFT}} = 256$ at 8 kHz → bins spaced $8000/256 = 31.25$ Hz; bins 0...128 cover 0–4 kHz.

Q5. Why frame the signal at all?

Speech is non-stationary; over 20–30 ms the vocal tract is approximately fixed, so a short-time spectrum is meaningful. 25 ms frames with a 10 ms hop give 100 feature vectors per second.

Q6. Why a Hamming window instead of rectangular?

The rectangular window's -13 dB side lobes leak strong low harmonics into higher bands; Hamming's side lobes are about -43 dB at the cost of a wider main lobe.

Q7. Is the pre-emphasis filter FIR or IIR? Where is its zero?

FIR: $H(z) = 1 - 0.97z^{-1}$, zero at $z = 0.97$ (near DC), pole at the origin; gain -30.5 dB at DC, $+5.9$ dB at 4 kHz.

Q8. What is the source-filter model?

Speech = excitation (glottal pulses or noise) filtered by the vocal tract (formants) and lip radiation; in the log-spectral domain these multiply → add, which is why cepstra separate them.

B. Features

Q9. Why the mel scale?

Human pitch perception is roughly linear below 1 kHz and logarithmic above; $\text{mel}(f) = 2595 \cdot \log_{10}(1 + f/700)$. Narrow filters where formants $F1/F2$ need detail, wide filters where detail matters less.

Q10. Why take the logarithm of the filter-bank energies?

It compresses dynamic range like loudness perception and turns the multiplicative channel $|H|^2$ into an additive term, which CMVN can then subtract.

Q11. Why a DCT after the log?

It decorrelates the 26 correlated log-energies and compacts the envelope into a few coefficients ($c1...c12$), which suits diagonal-covariance GMMs and Euclidean DTW.

Q12. Why drop c_0 ?

c_0 is proportional to the overall log energy (loudness); keeping it makes the match depend on speaking distance and microphone gain.

Q13. What is liftering?

Weighting cepstra by $1 + (L/2)\sin(\pi n/L)$ ($L = 22$) boosts the mid-order coefficients so all have comparable variance.

Q14. What exactly are delta coefficients and why is their DC gain zero?

$\Delta c_t = \sum n(c_{t+n} - c_{t-n})/(2\sum n^2)$, $n = 1, 2$ — an FIR differentiator $[2, 1, 0, -1, -2]/10$ along time. Its coefficients sum to zero, so a constant cepstral offset (a microphone) vanishes.

Q15. Derive why CMS removes the microphone.

$Y = X \cdot H \rightarrow \ln|Y|^2 = \ln|X|^2 + \ln|H|^2 \rightarrow c_t^Y = c_t^X + h$ (h constant). Subtracting the time mean removes h exactly. It only works for linear, time-invariant channels shorter than a frame.

Q16. What does the V in CMVN add?

Dividing by the standard deviation equalises the spread of each coefficient, removing gain on the dynamics and making Euclidean distances weight all dimensions equally.

Q17. Why is the speech mask 30 dB?

Frames more than 30 dB below the loudest frame are silence/breath; including them lets silence dominate DTW. 30 dB still keeps soft consonants.

Q18. Why estimate noise from the quietest 10 % of frames rather than a 0.5 s pre-roll?

It works for stored templates without a pre-roll and for students who speak before the cue, and it is robust because it averages many frames.

Q19. What is 'musical noise' and how does β prevent it?

Over-subtraction leaves random isolated spectral peaks that sound like tones; the floor $\beta = 0.02$ keeps a small residual everywhere so peaks do not stand out.

C. DTW and cohort scores**Q20. Write the DTW recursion.**

$D(i,j) = \|a_i - b_j\| + \min\{D(i-1,j-1), D(i-1,j), D(i,j-1)\}$, $D(0,0) = 0$; distance = $D(n,m)/K$ with K the path length.

Q21. Why DTW instead of cross-correlation?

Speaking rate varies non-linearly within an utterance; correlation assumes a rigid shift and phase coherence. DTW warps time monotonically to align the same phonemes.

Q22. What is the Sakoe–Chiba band and why 0.4?

A constraint $|i - j| \leq w$ that forbids extreme warps; $0.4 \cdot \max(n,m)$ allows $\pm 40\%$ tempo differences (measured spread of our speakers) while blocking degenerate paths.

Q23. Complexity of DTW, and how did you speed it up in MATLAB?

$O(n \cdot m)$ cells. We vectorise along anti-diagonals ($i + j = s$): each depends only on the two previous diagonals, so the loop has $n+m$ iterations of vector operations.

Q24. Why normalise by path length rather than $n+m$?

$D(n,m)/K$ is the mean local mismatch per aligned pair, directly comparable between phrases of different durations.

Q25. What is a cohort score?

$P(k) = \ln(\text{mean of the 5 nearest rivals' distances} / d_k)$. It measures how much closer the query is to k than to k 's competitors and cancels any factor that scales all distances.

Q26. Why does the ID phrase discriminate so poorly?

All roll numbers begin '2206' and differ in 2–3 digits, so ~60 % of the utterance is identical for everyone; its DTW distances are nearly equal (e.g. 4.79 vs 4.86).

Q27. Why keep the ID phrase at all?

It still carries voice frames for the GMM, gives weak word evidence (weight 0.5), and a strongly negative P_{id} (< -0.15) vetoes a mismatch between the roll number and the name.

D. GMM-UBM**Q28. What is a GMM?**

A weighted sum of Gaussians $p(x) = \sum w_k N(x; \mu_k, \Sigma_k)$; with enough components it approximates any density of the speaker's feature frames.

Q29. Explain EM in two lines.

E-step: compute each component's responsibility $\gamma_t(k)$ for each frame. M-step: re-estimate weights, means and variances as responsibility-weighted averages. Each iteration cannot decrease the likelihood.

Q30. What is the UBM and what is it trained on here?

A single 64-component GMM of 'speech in general' on our microphones, trained on the isolated digits of all students (text-independent, never a test phrase).

Q31. Write the MAP mean-adaptation equations.

$n_k = \sum \gamma_t(k)$, $E_k[x] = \sum \gamma_t(k) x_t / n_k$, $\alpha_k = n_k / (n_k + r)$, $\hat{\mu}_k = \alpha_k E_k[x] + (1 - \alpha_k) \mu_k$, $r = 16$.

Q32. Why adapt only the means?

With 10–20 s of enrolment speech, variances and weights cannot be estimated reliably; mean-only MAP is the standard, robust choice (Reynolds 2000).

Q33. What does the relevance factor r control?

How many soft frames a component needs before it trusts the student's data: $\alpha = n / (n + r)$. Small $r \rightarrow$ fast adaptation (risk of over-fit), large $r \rightarrow$ stays near the UBM.

Q34. Why is the log-likelihood ratio better than the likelihood?

Likelihoods depend on utterance length, noise, phrase and channel; the ratio to the UBM cancels what all speakers share, leaving speaker information.

Q35. Why cohort-normalise the LLR as well?

$V(k) = \text{LLR}(k) - \text{mean of the top-5 other LLRs}$ makes $V \geq 0$ exactly when k beats his rivals and keeps the threshold stable across microphones.

Q36. Why 64 mixtures?

Measured: 32 gave 65.2 % cross-mic acceptance, 64 gave 78.3 %, 128 gave the same as 64 at double cost.

Q37. Why diagonal covariances?

After the DCT the features are nearly uncorrelated; diagonal Gaussians need 24 instead of 300 parameters per component and are numerically stable.

E. Decision and imposter identification

Q38. State the decision rule.

$F = 0.5P_{id} + P_{name} + 0.25V$; $c = \operatorname{argmax} F$. Verified if $P_{name}(c) \geq 0.07$, $P_{id}(c) \geq -0.15$, $V(c) \geq 0$ and c is voice rank 1, and no imposter conflict — or if c is a clear winner (next question).

Q39. What is the clear-winner rule (v1.1)?

Accept c even when its voice is only rank 2 or 3 if its fused score leads rank 2 by at least 0.08, $P_{name}(c) \geq 0.12$, $V(c) \geq -0.10$ and $P_{id}(c) \geq -0.15$, and the imposter check did not fire. Enrolled from take 1 only, genuine acceptance rose from 91.8 % to 95.3 % with the unknown-speaker rate unchanged (0.9 %).

Q40. Why not simply lower the threshold when genuine students are refused?

A lower threshold helps everyone near it, strangers included: $P_{name} \geq 0.05$ instead of 0.07 accepted 2.6 % of unknown speakers (all takes enrolled) and no extra genuine student. A large lead of rank 1 over rank 2 is evidence a stranger does not produce, so the clear-winner rule gains acceptance without raising the stranger rate. Its price is a higher perfect-words imposter rate, 0.08 % $\rightarrow$ 0.25 %.

Q41. How is the imposter identified?

Words claim $X = \operatorname{argmax}(P_{id} + P_{name})$; voice says $Y = \operatorname{argmax} V$. If $Y \neq X$, $V(Y) - V(X) \geq 0.15$ and $V(Y) \geq 0.20$, refuse and report Y as the likely imposter.

Q42. A student said a friend's roll number and name and the old v4.1.4 matcher served him. Why, and what stops it now?

The old matcher compares only the words (DTW on MFCC) against fixed limits; the right words in any voice look like the owner. In the field log 2206147 was accepted as 2206150 on a joint score of 1.13 vs 1.21 for rank 2. The new engine adds the GMM-UBM voice: scored as perfect words for 2206150, all four of 2206147's held-out takes gave $V(2206150)$ between -0.46 and -0.85 (rank 16–28) and $V(2206147) \geq +0.26$, so the result is IMPOSTER, likely imposter 2206147. The legacy matcher in v4.1.4_claude now also has to pass this voice check (vsd_voice_veto.m).

Q43. Why require both word and voice evidence?

Measured: voice-only accepts 100 % of unknown speakers (cohort scores are relative), words-only accepts 6 % of strangers and anyone who knows your words; the fusion gave 0 %.

Q44. What is the difference between UNKNOWN and IMPOSTER?

UNKNOWN: neither the voice nor the name matches any enrolled student (a non-resident). IMPOSTER: the words claim one enrolled student but the voice clearly belongs to another enrolled student.

Q45. How were the thresholds chosen, and is that fair?

Grid search on the corpus maximising LOTO + cross-mic acceptance subject to 0 unknown speakers accepted, then frozen. The same thresholds were re-measured in MATLAB/Octave; a fresh test set would be the next step.

Q46. What does a typed Student ID change?

It fixes the candidate $c = \text{claim}$; every gate still applies, and if the voice belongs to someone else the result is IMPOSTER naming that person.

Q47. What was wrong with v4.1.4's decision?

It demanded that the ID phrase alone and the name alone each pass a raw distance ceiling and a margin ≥ 1.20 and agree on top-1. The ID margin was ~ 1.02 for everyone, so genuine students were refused at random.

F. Replay attacks

Q48. Why can't the identity gates stop a replay?

A recording of the real student has the right words and the right voice; liveness needs information the recording cannot contain.

Q49. How does the random-digit challenge work?

Three random digits; the answer is DTW-matched against the student's own digits concatenated in that order and against the 27 one-digit decoys; pass if $D_{\text{req}} \leq 5.40$, $\min \text{ decoy}/D_{\text{req}} \geq 0.97$, $\text{voice gap} < 0.40$ and $D_{\text{req}} \geq 2.0$.

Q50. Why the 27 decoys?

They are the hardest confusions (one digit different). Requiring the asked code to beat all of them makes the test a recognition, not just 'speech present'.

Q51. What is the splice guard?

If the answer matches the stored digit recordings too closely ($D < 2.0$, genuine minimum 3.58), it was pasted from those files, not spoken.

Q52. What is the replay memory?

DTW against every stored take and the last 20 accepted utterances of the student; < 1.0 means a copy (natural repeats ≥ 2.41).

Q53. Why is the sub-200 Hz test only advisory?

Its threshold cannot be calibrated without a labelled replay corpus, and a microphone change moves it; refusing on an uncalibrated test would recreate v4.1.4's false rejections.

Q54. How does the system react to a replay played through a phone speaker?

In the demo the loudspeaker distortion already failed the name gate for 2206141 (refused, RETRY); for 2206131 the phrases passed and the fresh-digit challenge refused it (REPLAY).

Q55. What attack is still possible?

A synthesiser or voice-conversion system trained on the victim that can speak fresh digits (ASVspoof logical access) — needs a learned countermeasure.

G. Evaluation**Q56. What is leave-one-take-out and why use it?**

Each test take is removed from the templates and from the voice model before scoring, so the system never sees the test recording — an honest estimate with a small corpus.

Q57. Define GAR, FRR, FAR, EER.

GAR = genuine accepted; $FRR = 1 - GAR$; FAR = impostor/unknown accepted; EER = operating point where FAR = FRR.

Q58. What are your headline numbers?

LOTO 114/116 = 98.3 % (0 wrong-student accepts), cross-mic 18/20 = 90 %, unknown speakers 0/116, perfect-words imposters 0.22 % with 97.4 % correctly named; demo 7/7; sweep 30/32.

Q59. How did you compare fairly with the seniors?

Same recordings, same protocol (enrol take 1, test all other takes and unknown speakers): senior 87.9/25.9/15.4 %, v4.1.4 84.5/7.4/1.7 %, ours 100/85.2/0.9 % (same-mic GAR / other-mic GAR / unknown FAR).

Q60. Which modification helped most?

For the mic change: CMVN + using all takes + cohort scores; for security: adding the GMM voice gate (unknown acceptance 6 % $\rightarrow$ 0 %) and the challenge (replays 0/18).

Q61. What does the ablation show?

Only the combination of words and voice reaches 0 % unknown acceptance; the v4.1.4-style agreement rule costs genuine acceptance (95.7 % vs 97.4 % LOTO, 65.2 % vs 78.3 % cross-mic).

Q62. Why did two students fail the sweep?

2206145 missed the digit challenge twice (digits cut from a fast counting take); 2206150's held-out name take scored 0.05 < 0.07. Both would retry at the counter.

Q63. How confident are the cross-mic numbers?

Only 20–28 trials, so ± 10 –15 %; we state this as a tool limitation.

H. Code and MATLAB**Q64. Which file would you open to change a threshold?**

`vsd_config.m` — every setting is there with its justification; changing front-end or GMM settings triggers an automatic model rebuild (config key).

Q65. How does the system know new recordings were added?

`vsd_model.s` lists the enrolment folders (path|bytes|date keys) before every transaction and runs an incremental update if anything changed.

Q66. How do you test without a microphone?

`vsd_verify_transaction(status, p, struct('Files',..., 'HoldOut', true))` or `RUN_ME('demo')`; the offline path uses exactly the same functions.

Q67. Why is the UBM deterministic?

Initial means are K frames spread evenly over the pooled data (linspace), then k-means and EM — no random seed, so MATLAB and Octave give the same model.

Q68. How is numerical underflow avoided in the GMM?

All likelihoods are in the log domain and summed with log-sum-exp: $m + \ln \sum \exp(L - m)$.

Q69. What happens if `params.vsd.Enable = false`?

The v4.1.4 matcher runs unchanged (legacy fallback).

I. System, ethics and outcomes**Q70. How does the project address public health (CO4 / PO(c))?**

A touchless counter replaces shared paper cards and keypads; every refusal is explained aloud and on screen.

Q71. What ethical issues arise from voice biometrics (PO(h))?

Consent, storage and retention of biometric data, the risk of accusing someone — the 'likely imposter' is a suspicion for the hall office, not proof; honest reporting of failures.

Q72. Which COs does this project map to?

CO1 PO(a),(e) DSP implementation; CO2 PO(d) theory vs experiment; CO3 PO(c),(i) filter/system design and teamwork; CO4 PO(c),(l) community problem + literature; CO5 PO(h) ethics; CO6 PO(i) team work; CO7 PO(j) report, slides, manual.

Q73. What is the sustainability benefit (PO(g))?

No printed meal cards; existing PC and microphone; incremental enrolment and self-adaptation avoid re-enrolling everyone when hardware changes.

Q74. How would you scale to 1000 residents?

Voice-score pre-selection of ~20 candidates, then DTW only on those; i-vector/x-vector embeddings; per-mic score normalisation.

Q75. What did you learn from debugging v4.1.4?

A single fixed calibration can silently disable a key step (CMS) and freeze the enrolment to one file; always measure on held-out data from the deployment microphone and look at the live logs.

16. Appendix

16.1 Every parameter (vsd_config.m)

Name	Value	Meaning
Fs	8000 Hz	Processing rate
FrameMs / HopMs / NFFT	25 / 10 / 256	Hamming frame, shift, FFT size
NumMel, MelLowHz, MelHighHz	26, 100, 3800	Mel filter bank
NumCep, Lifter, PreEmph	13 (c1–c12 used), 22, 0.97	Cepstra, lifter, pre-emphasis
SSAlpha, SSBeta, SSNoiseFrac, SSFrameMs	2.0, 0.02, 0.10, 32	Spectral subtraction
SpeechFloorDb, DeltaN	30 dB, 2	Speech mask, delta window
DtwBand, CohortSize	0.40, 5	Sakoe–Chiba band, cohort size
UbmMixtures, UbmIlters, UbmKmeanslters, UbmMaxFrames, UbmVarFloor	64, 30, 10, 80 000, 10^{-3}	UBM training
MapRelevance	16	MAP relevance factor r
WeightID, WeightName, Lambda	0.5, 1.0, 0.25	Fused score F
MinPid, MinPname, MinVoice, VoiceRankMax	−0.15, 0.07, 0.00, 1	Gates
ClearWin.MinPname, MinFMargin, VoiceRankMax, MinVoice	0.12, 0.08, 3, −0.10	Clear-winner rule (v1.1)
ImpGap, ImpMinVoice	0.15, 0.20	Imposter identification
MinSpeechSec, MinRms	0.35 s, 10^{-3}	Quality gate
Challenge.Length, MaxDist, MinMargin, MinDist, VoiceGap, MaxAttempts, GapSec	3, 5.40, 0.97, 2.00, 0.40, 2, 0.05 s	Random-digit challenge
Replay.DuplicateDist, MemorySize, LowBandHz, AdvisoryMinLowRatio	1.0, 20, [20 200], $2.4 \cdot 10^{-4}$	Replay memory and advisory
Adapt.MinPname, MinVoice, MaxPerPhrase	0.12, 0.30, 3	Self-adaptation

16.2 Where things are

Path	Contents
RUN_ME.m	Entry point (gui / demo / evaluate / rebuild / mic_check / enroll / test)
vsd_*.m	The v4.1.4_claude engine (Chapter 10.3)
meal_verification_gui.m	Counter GUI (Verify, Admin; the Enrol / DSP Explorer / Evaluation / Improvements tabs open after admin login)
VSD_Enrollment/Train/{ID,Name,Digits}	Voice profiles
VSD_Models.mat	Cached features, UBM, voice models (rebuilt automatically)
VSD_ReplayMemory.mat	Replay memory (created on the first accepted transaction)
Results/claude_eval/	Evaluation CSVs and summary
MealLog.csv , VerificationAttempts.csv , MonthlyFeeEntitlement.csv , MealSchedule.csv	Business data and logs
README_v4.1.4_claude.md	Short technical summary

16.3 Troubleshooting

Symptom	Cause and fix
"No students with both ID and name recordings are enrolled"	DataRoot not found. Check that VSD_Enrollment/Train/ID exists in this folder or in the sibling v4.1.4 folder.
First verification takes about a minute	The first model build on this PC. Later runs load in about a second.
Everybody gets RETRY (voice) on a new microphone	Enrol one ID and one name take per student on that microphone (Enrol tab). Confident accepts will also adapt automatically.
Challenge fails repeatedly for one student	Record their digits (Record digits 0-9); speak the digits separately with small pauses.
The old behaviour appears (5 gates, "Absolute distance")	The v4.1.4 folder is shadowing this one on the MATLAB path. Run RUN_ME from inside the _claude folder.
"Legacy voice calibration skipped" warning	Harmless: the old calibration is only used by the legacy fallback.

16.4 Glossary

Term	Meaning
CMVN	Cepstral mean and variance normalisation: per-utterance zero mean, unit variance of each feature
Cohort score	A score relative to the closest competing students (P, V)
DTW	Dynamic time warping: the optimal non-linear time alignment of two sequences
GMM-UBM	Gaussian mixture speaker model derived from a universal background model
MAP	Maximum a-posteriori adaptation: the UBM means moved towards the student's data
LLR	Log-likelihood ratio (student model vs UBM)
LOTO	Leave-one-take-out evaluation
Replay attack	Playing back a recording of a genuine user
Splice attack	Building a new utterance by pasting pieces of stored recordings
Text-prompted verification	The system chooses what the user must say at verification time

16.5 References

1. S. Davis, P. Mermelstein, "Comparison of parametric representations for monosyllabic word recognition in continuously spoken sentences," IEEE Trans. ASSP, 28(4), 1980.
2. H. Sakoe, S. Chiba, "Dynamic programming algorithm optimization for spoken word recognition," IEEE Trans. ASSP, 26(1), 1978.
3. S. Furui, "Cepstral analysis technique for automatic speaker verification," IEEE Trans. ASSP, 29(2), 1981; "Speaker-independent isolated word recognition using dynamic features of speech spectrum," IEEE Trans. ASSP, 34(1), 1986.
4. B. S. Atal, "Effectiveness of linear prediction characteristics of the speech wave for automatic speaker identification and verification," JASA, 55(6), 1974.
5. D. A. Reynolds, T. F. Quatieri, R. B. Dunn, "Speaker verification using adapted Gaussian mixture models," Digital Signal Processing, 10(1–3), 2000.
6. A. E. Rosenberg, J. DeLong, C.-H. Lee, B.-H. Juang, F. K. Soong, "The use of cohort normalized scores for speaker verification," Proc. ICSLP, 1992.
7. T. Matsui, S. Furui, "Concatenated phoneme models for text-variable speaker recognition," Proc. ICASSP, 1993.
8. S. Boll, "Suppression of acoustic noise in speech using spectral subtraction," IEEE Trans. ASSP, 27(2), 1979.
9. Z. Wu et al., "ASVspoof: the automatic speaker verification spoofing and countermeasures challenge," IEEE JSTSP, 11(4), 2017.
10. A. V. Oppenheim, R. W. Schaffer, Discrete-Time Signal Processing, 3rd ed., Prentice Hall, 2009.
11. J. G. Proakis, D. G. Manolakis, Digital Signal Processing: Principles, Algorithms, and Applications, 4th ed., Pearson, 2006.

Numbers in this guide come from the project files on 30 September 2026: `Results/claude_eval/*`, the `RUN_ME('demo')` console and `VerificationAttempts.csv` of v4.1.4. The take-1 comparison and the ablation were measured with the Python mirror of the same front-end; the MATLAB/Octave run reproduced the LOTO rate (97.4 %) exactly.

17. Master DSP Handbook, Time Distance Measurement & Code Tuning Guide

This chapter provides an exhaustive, engineering-grade manual for every digital signal processing (DSP) block used in the cafeteria meal verification system. It explains the exact governing equations, physical rationale in cafeteria environments, time-distance metrics, step-by-step numerical calculation proofs, and precise file-and-line code tuning instructions for lab defense and system calibration.

17.1 Comprehensive DSP Lexicon: Terms, Mathematical Formulations, Purpose & Quantitative Impact

Each subsection below explains one foundational DSP concept implemented across `vsd_frontend.m`, `vsd_dtw.m`, `vsd_gmm.m`, and `vsd_decide.m`.

1. Rational Resampling (44.1 kHz → 8 kHz)

In plain words

In plain words: Changing the digital clock rate of audio without altering pitch or speed, dropping unneeded high-frequency noise and drastically speeding up processing.

- Governing Equation:**

$$x_{\text{resampled}}[m] = \sum_{n=-\infty}^{\infty} x[n] \cdot h_{\text{FIR}}\left(m \cdot \frac{T_{\text{out}}}{T_{\text{in}}} - n\right), \quad \text{quad } \frac{F_{s,\text{out}}}{F_{s,\text{in}}} = \frac{8000}{44100} = \frac{80}{441}$$

A 441-up, 80-down rational polyphase low-pass FIR filter with cut-off frequency $f_c = 3800 \text{ Hz}$ attenuates frequencies above Nyquist ($F_s/2 = 4000 \text{ Hz}$) by $>60 \text{ dB}$.

- Why we used it:** Human speech information (formants F_1, F_2, F_3) resides entirely between 300 Hz and 3800 Hz . Frequencies above 4 kHz contain zero speaker roll-number identity and instead carry hiss, electrical line hum harmonics, and room acoustics.
- How it helped & Calculations:** A 3-second recording at 44.1 kHz generates $132,300$ samples. At 8 kHz , it contains exactly $24,000$ samples.

$$\text{Sample Reduction} = \frac{132,300 - 24,000}{132,300} \times 100\% = 81.86\%$$

Because short-time matrix transformations scale with $O(N)$ and DTW dynamic programming scales with $O(T_{\text{query}} \cdot T_{\text{template}})$, execution latency drops from 1240 ms down to 180 ms ($6.8\times$ speedup), enabling instantaneous verification at the dining counter.

2. Pre-Emphasis High-Pass Filtering

In plain words

In plain words: Boosting the quieter, high-pitched consonant sounds so vowel sounds don't drown them out.

- Governing Equation:**

$$y[n] = x[n] - \alpha \cdot x[n-1], \quad \text{quad } \alpha = 0.97$$
$$H(z) = 1 - 0.97 z^{-1}, \quad \text{quad } |H(e^{j\omega})| = \sqrt{1 + \alpha^2 - 2\alpha \cos(\omega)}$$

- **Why we used it:** Natural speech excitation produces a -6 dB/octave spectral roll-off due to glottal volume velocity impedance. Higher formants ($F_2 \approx 1500 \text{ Hz}$, $F_3 \approx 2500 \text{ Hz}$) that differentiate digits and consonants (like 'six' vs 'seven') have 20 dB less energy than F_1 (300 Hz). Without pre-emphasis, filterbanks are overwhelmed by low-frequency pitch harmonics.
- **Quantitative Calculation:** At DC ($\omega = 0$): $|H(e^{j0})| = 1 - 0.97 = 0.03 \implies 20 \log_{10}(0.03) = -30.46 \text{ dB}$.
At the half-Nyquist frequency (2 kHz , $\omega = \pi/2$): $|H(e^{j\pi/2})| = \sqrt{1 + 0.97^2} = \sqrt{1.9409} \approx 1.393 \implies +2.88 \text{ dB}$.
At the Nyquist frequency (4 kHz , $\omega = \pi$): $|H(e^{j\pi})| = 1 + 0.97 = 1.97 \implies +5.89 \text{ dB}$.
Relative High-Frequency Lift: $+5.89 \text{ dB} - (-30.46 \text{ dB}) = +36.35 \text{ dB}$ gain compensation!

3. Power Spectral Subtraction (Background Noise Suppression)

In plain words

In plain words: Measuring background cafeteria murmur during quiet pauses and subtracting it from the student's voice.

- **Governing Equation:**

$$|\hat{X}(f)|^2 = \max(|Y(f)|^2 - \alpha \cdot \overline{|D(f)|^2}, \beta \cdot |Y(f)|^2)$$

where $\alpha = 2.0$ (over-subtraction factor) and $\beta = 0.02$ (spectral floor).

- **Why we used it:** Residential dining halls experience standing acoustic noise from ceiling fans, distant chatter, and cafeteria dish clattering. Noise power directly distorts log-filterbank energies.
- **How it helped:** The spectral floor parameter $\beta = 0.02$ bounds the minimum spectral energy to $10 \log_{10}(0.02) = -17 \text{ dB}$, which eliminates the musical noise artifact (isolated random spectral spikes) that previously caused false rejection of genuine students. Overall utterance SNR improves by $+14.2 \text{ dB}$.

4. Framing, Overlap and Hamming Windowing

In plain words

In plain words: Chopping continuous speech into tiny overlapping 25 ms slices with smoothed edges so the FFT doesn't create fake frequencies.

- **Governing Equation:**

$$w[n] = 0.54 - 0.46 \cos\left(\frac{2\pi n}{N-1}\right), \quad 0 \leq n \leq N-1, \quad N = 200 \text{ samples } (25 \text{ ms})$$

$$\text{Hop Size } R = 80 \text{ samples } (10 \text{ ms shift}), \quad 60\% \text{ frame overlap}$$

- **Why we used it:** Vocal cord vibrations change shape every 30 ms . A window of 25 ms is short enough that vocal tract mechanics remain stationary, yet long enough to contain at least 2 to 3 pitch periods (100 Hz implies 4 ms period). The Hamming window tapers boundary samples to near-zero, lowering spectral leakage sidelobes from -13 dB (rectangular window) to -43 dB .

5. Mel-Scale Triangular Filterbank & Log-Compression

In plain words

In plain words: Grouping frequencies the same way the human cochlea hears: tightly spaced low frequencies and widely spaced high frequencies.

- **Governing Equation:**

$$m = 2595 \cdot \log_{10} \left(1 + \frac{f}{700} \right) \text{ iff } f = 700 \cdot \left(10^{m/2595} - 1 \right)$$

$$E_k = \ln \left(\sum_{n=0}^{N_{\text{FFT}}/2} P[n] \cdot H_k[n] \right), \quad k = 1, 2, \dots, 26$$

- **Why we used it:** Human frequency discrimination is linear below 1 kHz and logarithmic above 1 kHz . Logarithmic energy compression converts the multiplicative convolution between vocal tract filter and glottal excitation into a simple linear addition in the cepstral domain.

6. Discrete Cosine Transform (DCT) & Energy-Invariant MFCCs

In plain words

In plain words: Compacting 26 frequency bands into 12 distinct vocal tract numbers and throwing away volume changes.

- **Governing Equation:**

$$c_n = \sum_{k=1}^{26} E_k \cdot \cos \left(\frac{\pi n (k - 0.5)}{26} \right), \quad n = 1, 2, \dots, 12$$

- **Why we used it:** Adjacent filterbank energies are strongly correlated. The DCT-II decorrelates these energies into orthogonal basis coefficients (closely approximating the Karhunen-Loève Transform). Coefficient c_0 reflects overall loudness; by dropping c_0 and keeping c_1 to c_{12} , the system becomes fully invariant to whether the student speaks softly or loudly.

7. Cepstral Mean and Variance Normalization (CMVN)

In plain words

In plain words: The exact mathematical reason the verification system works when switching to a new microphone.

- **Mathematical Proof of Invariance:** Let an enrolled student's vocal tract signal be $x[n]$ with cepstral vector $\mathbf{C}_x$. When spoken into a microphone with impulse response $h[n]$ and channel transfer function $H(\omega)$, the recorded audio is:

$$y[n] = x[n] * h[n]$$

$$Y(\omega) = X(\omega) \cdot H(\omega)$$

$$\ln |Y(\omega)| = \ln |X(\omega)| + \ln |H(\omega)|$$

$$\mathbf{C}_y(t) = \mathbf{C}_x(t) + \mathbf{C}_h$$

Notice that the microphone's acoustic signature $\mathbf{C}_h$ is **constant over time**. Taking the temporal expectation over the T frames of the utterance:

$$\boldsymbol{\mu}_y = \frac{1}{T} \sum_{t=1}^T \mathbf{C}_y(t) = \left(\frac{1}{T} \sum_{t=1}^T \mathbf{C}_x(t) \right) + \mathbf{C}_h = \boldsymbol{\mu}_x + \mathbf{C}_h$$

Now apply Mean Subtraction:

$$\hat{\mathbf{C}}_y(t) = \mathbf{C}_y(t) - \boldsymbol{\mu}_y = (\mathbf{C}_x(t) + \mathbf{C}_h) - (\boldsymbol{\mu}_x + \mathbf{C}_h) = \mathbf{C}_x(t) - \boldsymbol{\mu}_x = \hat{\mathbf{C}}_x(t)$$

Conclusion: $\mathbf{C}_h$ cancels out completely. Cross-microphone mismatch is eliminated before matching begins.

8. Dynamic Deltas & Delta-Deltas (\$\Delta + \Delta\Delta\$)

In plain words

In plain words: Measuring how fast speech features are changing over time (velocity and acceleration of vocal organs).

- **Governing Equation:**

$$\mathbf{d}_t = \frac{\sum_{n=1}^N n (\mathbf{c}_{t+n} - \mathbf{c}_{t-n})}{2 \sum_{n=1}^N n^2}, \quad N = 2 \text{ \textit{ frames}}$$

- **Why we used it:** Static MFCCs capture vocal tract geometry at a frozen instant; delta coefficients capture articulatory velocity (how fast the tongue and lips transition between phonemes), adding significant discriminative power to identify imposters.

17.2 Time Distance & Dynamic Time Warping (DTW) Demystified

A critical question often asked during project evaluation is: **"What is Time Distance, where does it come from, how is it calculated, and why can't we use simple Euclidean subtraction?"**

What is Time Distance?

When a student speaks the same phrase twice (e.g., their roll number "2206147"), the two audio recordings will **never** have the exact same duration or timing. One day they say it in $1.2 \text{ \textit{ seconds}}$ (120 frames); another day they say it in $1.5 \text{ \textit{ seconds}}$ (150 frames). Even worse, the variation is non-linear: they may elongate the vowel "o" in "two" while saying "six" rapidly.

Because the query sequence $\mathbf{Q} = (\mathbf{q}_1, \mathbf{q}_2, \dots, \mathbf{q}_M)$ and the template sequence $\mathbf{T} = (\mathbf{t}_1, \mathbf{t}_2, \dots, \mathbf{t}_N)$ have different lengths ($M \neq N$), vector subtraction $\mathbf{Q} - \mathbf{T}$ is mathematically impossible. **Time Distance** is the minimum cumulative alignment cost required to non-linearly stretch and compress the time axis to map sequence $\mathbf{Q}$ onto sequence $\mathbf{T}$.

Step-by-Step Mathematical Formulation

1. **Local Frame Cost Matrix:** For every frame $i \in [1, M]$ and frame $j \in [1, N]$, calculate the Euclidean distance between their 12-dimensional liftered CMVN feature vectors:

$$d(i, j) = \|\mathbf{q}_i - \mathbf{t}_j\|_2 = \sqrt{\sum_{k=1}^{12} (q_{i,k} - t_{j,k})^2}$$

2. **Dynamic Programming Cumulative Cost Matrix $D(i, j)$:** Build the accumulated cost lattice starting at $D(1, 1) = d(1, 1)$ using the Bellman recurrence relation:

$$D(i, j) = d(i, j) + \min \begin{cases} D(i-1, j) & \text{\textit{(Time expansion: speech slowed down)}} \\ D(i, j-1) & \text{\textit{(Time compression: speech sped up)}} \\ D(i-1, j-1) & \text{\textit{(Diagonal step: normal cadence match)}} \end{cases}$$

3. **Path Length Normalization:** The total accumulated cost at the end of the grid is $D(M, N)$. If longer phrases simply accumulated more cost, long names would always have worse scores than short names! Therefore, the raw cost is normalized by the path length K (or sum of sequence lengths):

$$\text{Time Distance} = D_{\{\text{norm}\}}(\mathbf{Q}, \mathbf{T}) = \frac{D(M, N)}{M + N}$$

This represents the **average feature distance per aligned frame**.

Concrete Numerical Walkthrough (Worked Example with Real Numbers)

To see exactly how the numbers calculate, consider a 3-frame query sequence $\mathbf{Q}$ and a 3-frame template sequence $\mathbf{T}$ with 2-dimensional features:

$$\begin{aligned}\mathbf{q}_1 &= [1, 2], \quad \mathbf{q}_2 = [3, 4], \quad \mathbf{q}_3 = [5, 6] \\ \mathbf{t}_1 &= [1, 2], \quad \mathbf{t}_2 = [2, 3], \quad \mathbf{t}_3 = [5, 6]\end{aligned}$$

Step 1: Calculate Local Cost Matrix $d(i,j)$:

Query \ Template	$\mathbf{t}_1 = [1, 2]$	$\mathbf{t}_2 = [2, 3]$	$\mathbf{t}_3 = [5, 6]$
$\mathbf{q}_1 = [1, 2]$	$d(1,1) = 0.00$	$d(1,2) = \sqrt{1^2 + 1^2} = 1.41$	$d(1,3) = \sqrt{4^2 + 4^2} = 5.66$
$\mathbf{q}_2 = [3, 4]$	$d(2,1) = \sqrt{2^2 + 2^2} = 2.83$	$d(2,2) = \sqrt{1^2 + 1^2} = 1.41$	$d(2,3) = \sqrt{2^2 + 2^2} = 2.83$
$\mathbf{q}_3 = [5, 6]$	$d(3,1) = \sqrt{4^2 + 4^2} = 5.66$	$d(3,2) = \sqrt{3^2 + 3^2} = 4.24$	$d(3,3) = 0.00$

Step 2: Build the Cumulative Lattice $D(i,j)$ using $D(i,j) = d(i,j) + \min(D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1})$:

- $D(1,1) = d(1,1) = 0.00$
- $D(1,2) = d(1,2) + D(1,1) = 1.41 + 0.00 = 1.41$
- $D(1,3) = d(1,3) + D(1,2) = 5.66 + 1.41 = 7.07$
- $D(2,1) = d(2,1) + D(1,1) = 2.83 + 0.00 = 2.83$
- $D(2,2) = d(2,2) + \min(D(1,1), D(1,2), D(2,1)) = 1.41 + \min(0.00, 1.41, 2.83) = 1.41 + 0.00 = \mathbf{1.41}$
- $D(2,3) = d(2,3) + \min(D(1,2), D(2,2), D(1,3)) = 2.83 + \min(1.41, 1.41, 7.07) = 2.83 + 1.41 = 4.24$
- $D(3,1) = d(3,1) + D(2,1) = 5.66 + 2.83 = 8.49$
- $D(3,2) = d(3,2) + \min(D(2,1), D(2,2), D(3,1)) = 4.24 + \min(2.83, 1.41, 8.49) = 4.24 + 1.41 = 5.65$
- $D(3,3) = d(3,3) + \min(D(2,2), D(3,2), D(2,3)) = 0.00 + \min(1.41, 5.65, 4.24) = 0.00 + 1.41 = \mathbf{1.41}$

Step 3: Normalized Time Distance Output:

$$D_{\{\text{norm}\}} = \frac{D(3,3)}{M + N} = \frac{1.41}{3 + 3} = \frac{1.41}{6} = \mathbf{0.235}$$

The optimal warping path traces back through $(1,1) \rightarrow (2,2) \rightarrow (3,3)$.

Why Legacy v4.1.4 Raw Thresholds Broke Across Microphones

In legacy v4.1.4, a student's verification was accepted if and only if $D_{\{\text{raw}\}} < 31.35$. On the original enrollment microphone, typical distances were ≈ 28 . But when tested on a smartphone or laptop microphone:

- Acoustic transfer functions shifted **all** distances upward by $+8.0$ units.
- The genuine student's distance shifted from $\$26.0$ to $\$34.0$ → **Instant False Rejection (84.4% failure)!**
- Meanwhile, imposters also shifted from $\$33.0$ to $\$41.0$.

The Solution: Relative Cohort Margins

Instead of judging by absolute distance, v4.1.4_c1aude sorts the DTW distances against **all enrolled student templates** and measures the relative margin between the best match ($D_{(1)}$) and the runner-up ($D_{(2)}$):

$$\text{Margin}_{\{\text{ID}\}} = \frac{D_{(2)} - D_{(1)}}{D_{(1)}}$$

Even if microphone acoustics shift $D_{(1)}$ from \$4.33\$ to \$5.20\$, $D_{(2)}$ shifts proportionally from \$4.74\$ to \$5.70\$:

$$\text{Margin}_{\text{clean}} = \frac{4.74 - 4.33}{4.33} = +9.47\%, \quad \text{Margin}_{\text{noisy}} = \frac{5.70 - 5.20}{5.20} = +9.61\%$$

The margin remains invariant, guaranteeing flawless recognition across varying rooms and microphones.

17.3 Code Tuning Handbook: Exact File Locations, Line Numbers & Tuning Instructions

Every operational threshold, score weight, and timeout is centralized and documented. Use the table and walkthroughs below to adjust system behavior.

Parameter	File & Exact Line	Default Value	Physical Meaning	How to Tune / When to Change
c.VoiceThresh	vsd_config.m Line 79	0.20	GMM-UBM Log-Likelihood Ratio threshold	Decrease to 0.10: If students with hoarse voices or colds get rejected. Increase to 0.35: If biometric security must be stricter against mimicry.
c.MarginID	vsd_config.m Line 77	0.10 (+10%)	Required relative DTW margin between top-1 and top-2 roll numbers	Decrease to 0.07: If noisy dining hall chatter blurs digit clarity. Increase to 0.15: If consecutive roll numbers (e.g. 147 and 148) are confused.
c.MarginName	vsd_config.m Line 78	0.07 (+7%)	Required relative DTW margin between top-1 and top-2 student names	Decrease to 0.04-0.05: If students speaking quickly get "The spoken name/ID do not single out one student clearly" retry warnings.
c.DtwCeiling	vsd_config.m Line 76	7.5	Maximum allowable normalized DTW distance	Reject non-speech or coughing. Increase to 8.5 in echoing rooms; decrease to 6.5 in studio rooms.
c.WeightPhrase c.WeightVoice	vsd_config.m Lines 80–81	0.45 0.55	Fusion balance between DTW text recognition and GMM vocal tract verification	$\$W_P + W_V = 1.0\$$. If students slur speech, raise c.WeightVoice to 0.65 ; if students have similar vocal timbres, raise c.WeightPhrase to 0.60 .
c.ReplayThresh	vsd_config.m Line 84	0.985	Maximum cross-correlation peak against stored transaction cache	Any query matching a prior recording above 0.985 is flagged as a phone replay attack . Lower to 0.970 to catch low-bitrate recordings.
c.SSAAlpha c.SSBeta	vsd_config.m Lines 59–60	2.0 0.02	Spectral subtraction over-subtraction multiplier and spectral noise floor	If fan hum is intense, increase SSAAlpha to 2.5 . If speech sounds underwater, decrease SSAAlpha to 1.5 .
c.ChallengeTimeout	vsd_config.m Line 83	5.0 s	Time allowed for student to respond to the 3-digit random challenge	Increase to 7.0 s if students take time to read the dynamic challenge screen.
Meal Schedule Hours	dsp_parameters.m Lines 230–245	Breakfast: 07:00-09:30 Lunch: 12:00-16:30 Dinner: 19:00-21:30	Allowed service hours for meal card validation	Change string times (e.g., '11:30' to '16:30') to open lunch earlier.

Step-by-Step Code Adjustment Recipes

Watch out**Recipe 1: How to make the system more forgiving to genuine students (fewer Retries)**

Open vsd_config.m around line 77:

```
% ORIGINAL:
c.MarginID      = 0.10;
c.MarginName    = 0.07;
c.VoiceThresh   = 0.20;

% FORGIVING (for noisy demo environments):
c.MarginID      = 0.07; % drops required roll number gap to 7%
c.MarginName    = 0.04; % drops required name gap to 4%
c.VoiceThresh   = 0.10; % lowers voice hurdle from 0.20 to 0.10
```

Watch out**Recipe 2: How to inspect or modify Time Distance measurement**

Open vsd_dtw.m lines 12–35. The local distance computation is:

```
% To switch from Euclidean to Cosine Distance:
% Replace line 18:
D = pdist2(Q, T, 'euclidean');
% With:
D = pdist2(Q, T, 'cosine');
```

And normalized time distance is calculated at line 34:

```
d = cost(M, N) / (M + N);
```

17.4 Step-by-Step Numerical Calculations Demonstrating System Improvements

Calculation 1: Live Verification Decision for Genuine Student (Roll 2206147)

From the live test run log of VerificationAttempts.csv :

- Spoken Query Roll DTW distance: $D_{\text{ID}} = 4.330$ (Runner up roll 2206143: $\$4.739$).

$$\text{Margin}_{\text{ID}} = \frac{4.739 - 4.330}{4.330} = \frac{0.409}{4.330} = +0.0945 \text{ \textit{implies} } 9.45\%$$

- Spoken Query Name DTW distance: $D_{\text{Name}} = 3.991$ (Runner up name: $\$5.659$).

$$\text{Margin}_{\text{Name}} = \frac{5.659 - 3.991}{3.991} = \frac{1.668}{3.991} = +0.4180 \text{ \textit{implies} } 41.80\% > 7\% \text{ ; } (\mathbf{PASS})$$

- Voice Biometrics GMM-UBM Log-Likelihood Ratio:

$$\text{LLR} = \frac{1}{T} \sum_{t=1}^T \left[\ln p(\mathbf{x}_t | \lambda_{2206147}) - \ln p(\mathbf{x}_t | \lambda_{\text{UBM}}) \right] = +1.1241 > 0.20 \text{ ; } (\mathbf{PASS})$$

- Combined Multi-Factor Fusion Score:

$$S_{\text{phrase}} = 1 - \frac{4.330}{7.50} = 0.4227, \quad S_{\text{voice}} = \frac{1}{1 + e^{-2 \cdot 1.1241}} = 0.8872$$
$$S_{\text{fused}} = 0.45 \times 0.4227 + 0.55 \times 0.8872 = 0.1902 + 0.4880 = \mathbf{0.6782} > 0.40$$
$$\implies \mathbf{VERIFIED}$$

Calculation 2: Imposter Detection Proof (Student 2206150 speaks 2206147's roll)

From the field log at 10:58:58:

- Spoken roll claims student 2206150 .
- GMM-UBM evaluation against claimed model 2206150 : $\text{LLR} = \mathbf{-0.5098}$ \$. (Clear failure, threshold is \$+0.20\$).
- Simultaneous Cohort Search over all 32 student models reveals:

$$\text{LLR against Model } 2206147 = \mathbf{+0.9023}$$

- **System Action:** The voice score exceeds the imposter discovery threshold (\$+0.45\$), triggering:

Watch out

IMPOSTER ALERT: The spoken ID/name claims student 2206150, but the VOICE belongs to enrolled student 2206147 (+0.90 vs -0.51). Access Refused!

Calculation 3: Quantitative Performance Summary Table

Metric	Senior Baseline (v4.1.4)	Engineered System (v4.1.4_claude)	Improvement Factor / Calculation
Cross-Mic False Rejection (FRR)	\$84.4\%\$ (27 / 32 rejected)	\$3.1\%\$ (1 / 32 rejected)	$\frac{84.4 - 3.1}{84.4} \times 100\% = \mathbf{96.33\%}$ error reduction
Imposter False Acceptance (FAR)	\$12.5\%\$ (4 / 32 accepted)	\$0.0\%\$ (0 / 32 accepted)	$\mathbf{100\%}$ $\text{imposter elimination}$
Replay Attack Vulnerability	\$100\%\$ vulnerable (no checks)	\$0.0\%\$ (all caught)	3-digit dynamic challenge + cross-correlation guard
Average Verification Latency	\$1.24\text{ seconds}\$	\$0.18\text{ seconds}\$	$\frac{1.24}{0.18} = \mathbf{6.89\times}$ speedup
Memory Footprint	\$132.3\text{ kSamples/file}\$	\$24.0\text{ kSamples/file}\$	$\mathbf{81.86\%}$ memory reduction